BONDING IN MOLECULES

General Chemistry I, Lecture Series 9

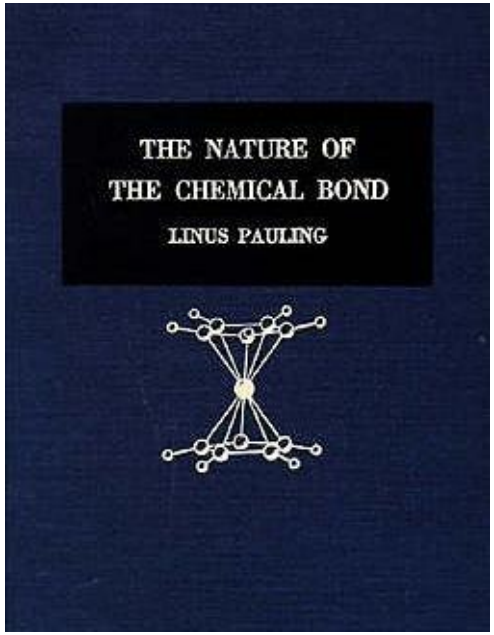
Qixi Mi

Reading:

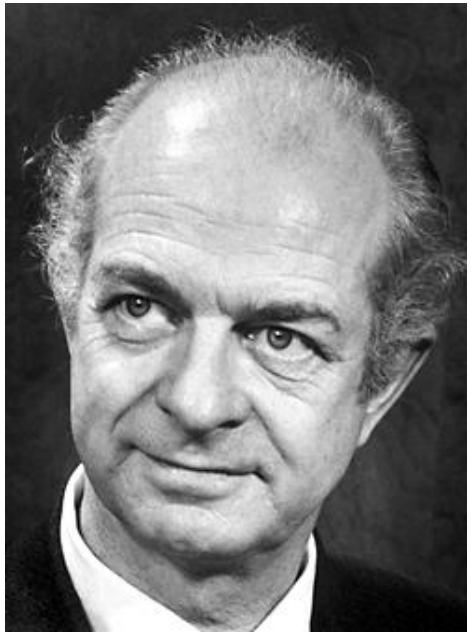
OGB8 §6, YY §§3.2–3.5

© Qixi Mi 2014–25, except pictures from the Internet.

Chemistry in the 20th Century



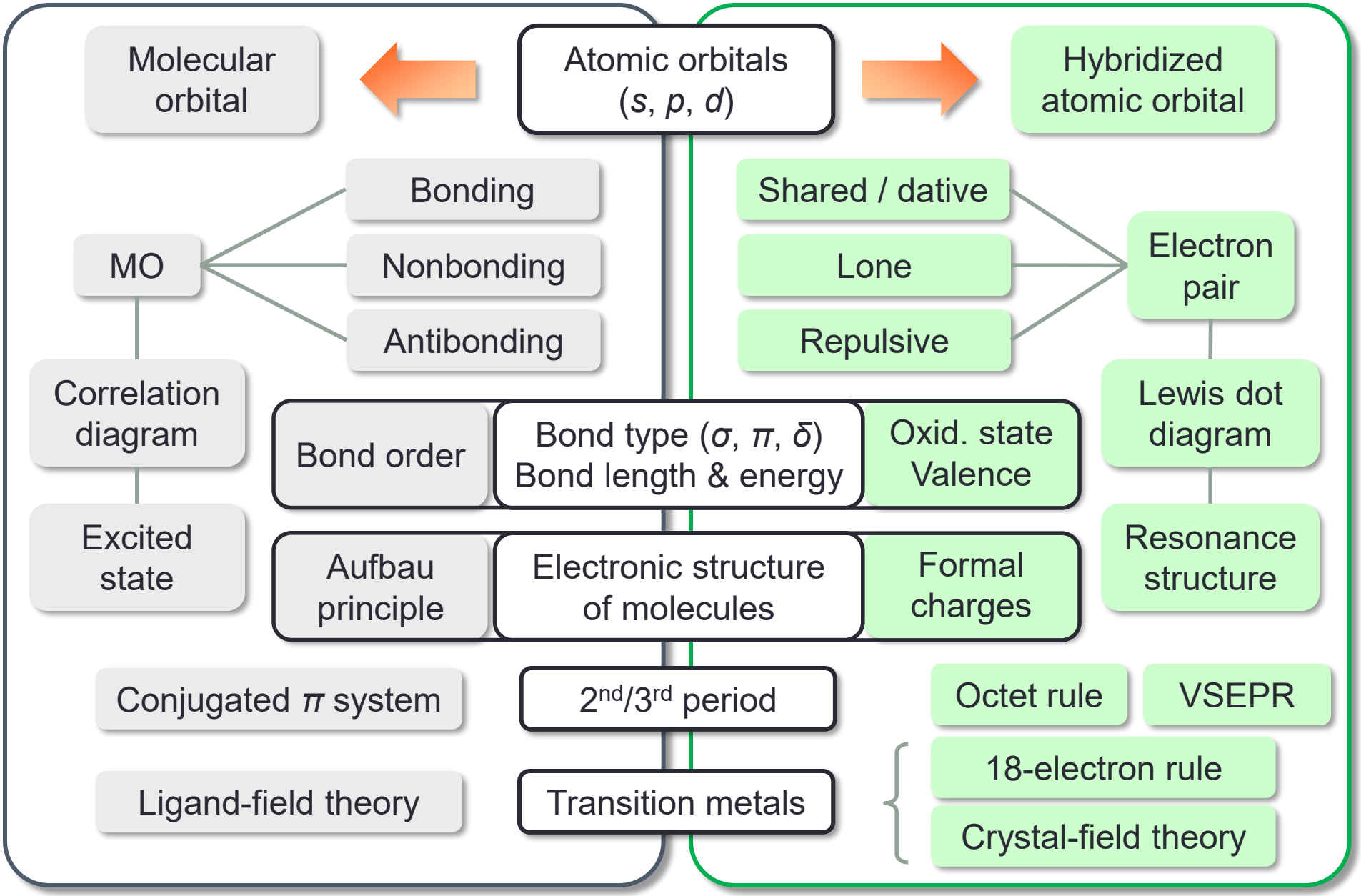
The Nature of the Chemical Bond (1939)



Linus Pauling (Caltech, 1901–1994)

Molecular Orbital (MO)

Valence Bond (VB) / Hybridization



The Nature of the Chemical Bond

$$H + H \longrightarrow H{-}H$$

$$He + He \not\longrightarrow He{-}He \quad \text{or} \quad He{=}He$$

$$H + H^{+} \longrightarrow (H{\cdots}H)^{+}$$

$$He + He^{+} \longrightarrow (He{\cdots}He)^{+}$$

$$Kr^{*} + F \longrightarrow (Kr{\cdots}F)^{*}$$

Excimer
受激准分子

From Classical to Modern Chemistry

- The way of thinking

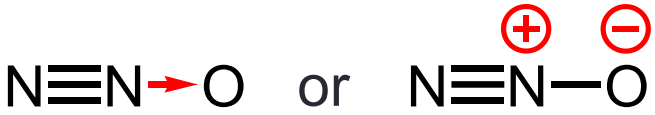
Valence
化合价

→

Orbital 轨道
Bonding 成键

- The language

$\overset{+1}{N}_2\overset{-2}{O}$ →



- Capability to analyze unfamiliar systems under theoretical guidance

Outline

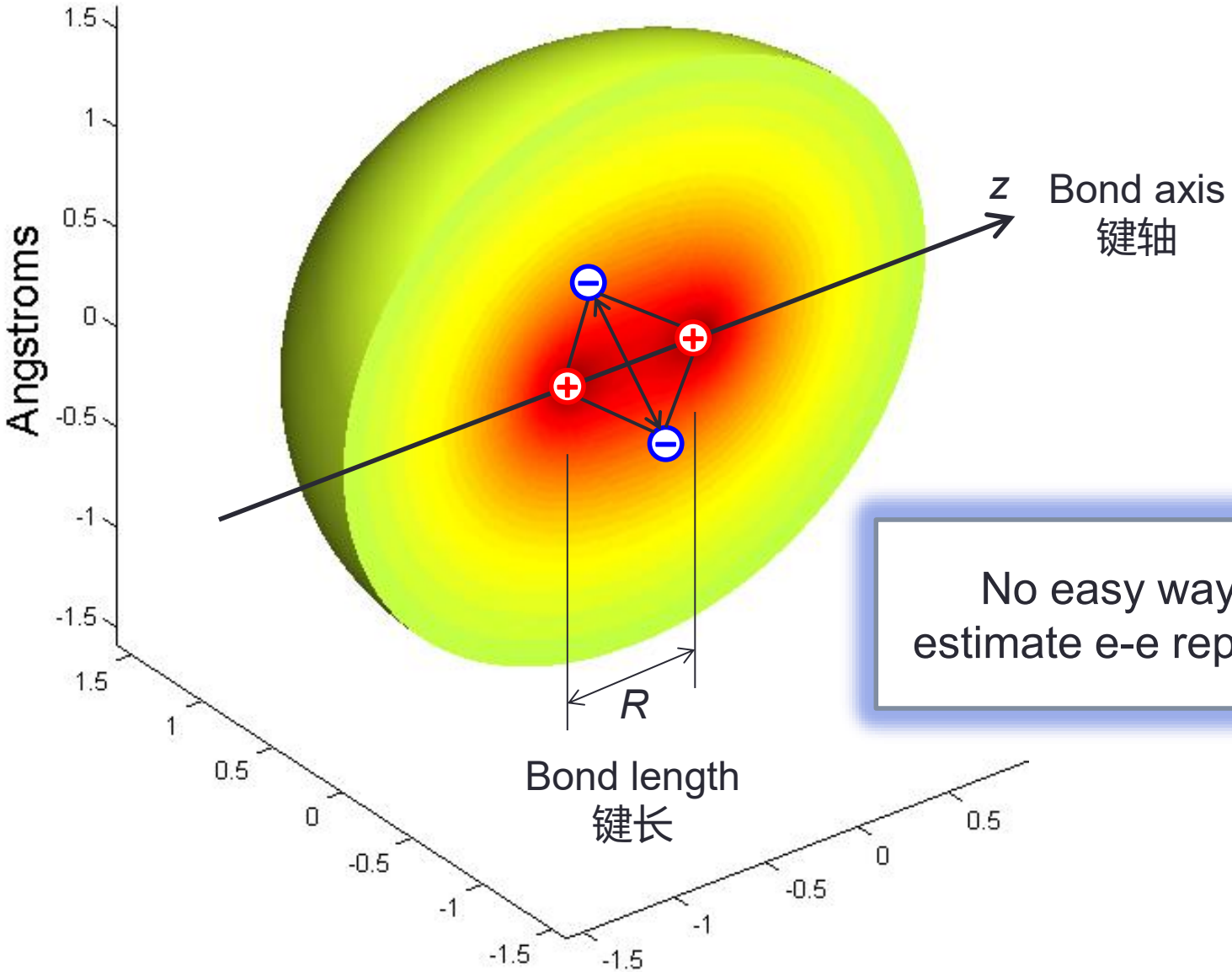
- Overview

- Molecular Orbital Theory (1927)
  - LCAO (1929)
  - Orbital correlation diagram

- Types of bonding
  - σ, π bonds
  - Bonding, antibonding, nonbonding
  - Bond order

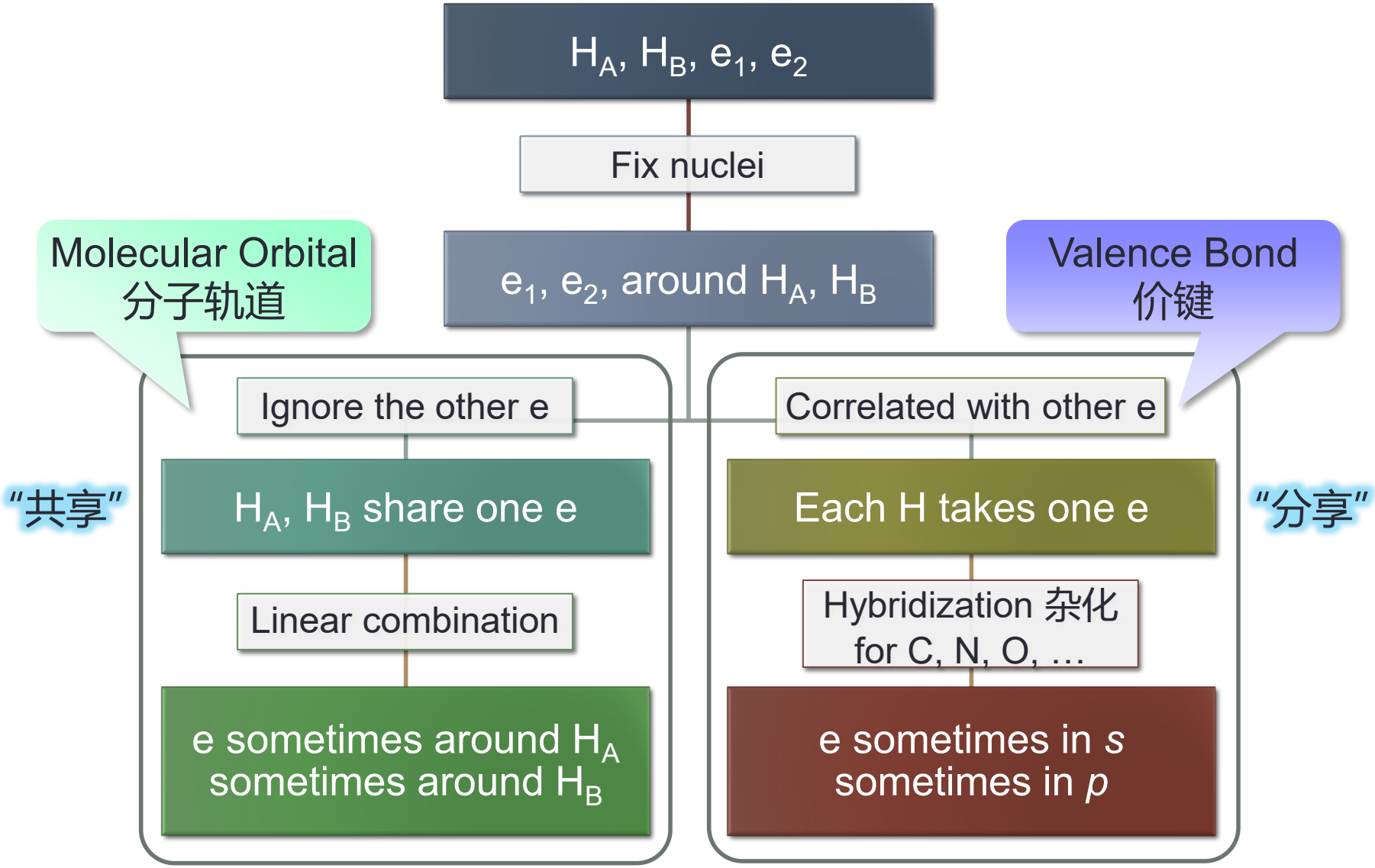
- Valence Bond Theory (1927)
  - Hybridization (1930)

H_2: the Simplest Molecule



No easy way to estimate e-e repulsion

H_2: Levels of Approximation



Characteristics of the Two Theories

Molecular Orbital (MO)

- Ignores e-e repulsion
- Electrons wander around
- For orbitals that are
  - Delocalized 离域
  - 'Fat'
  - Singly occupied

Valence Bond (VB)

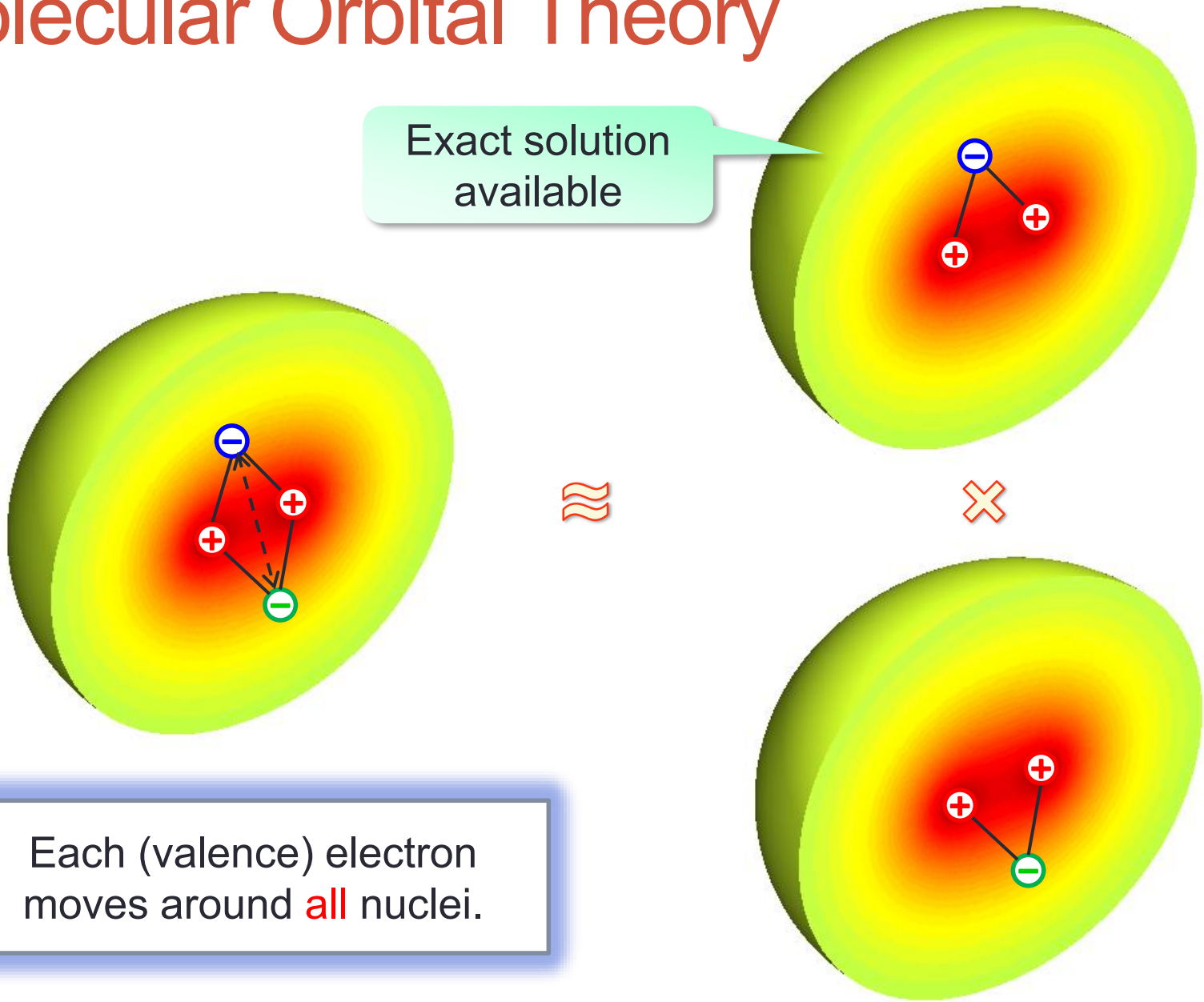
- Exaggerates e-e repulsion
- Electrons adhere to the nuclei
- For orbitals that are
  - Localized 定域/局域
  - 'Tight'
  - Strong

Outline

- Overview
- Molecular Orbital Theory (1927)
  - LCAO (1929)
  - Orbital correlation diagram
- Types of bonding
  - σ, π bonds
  - Bonding, antibonding, nonbonding
  - Bond order
- Valence Bond Theory (1927)
  - Hybridization (1930)

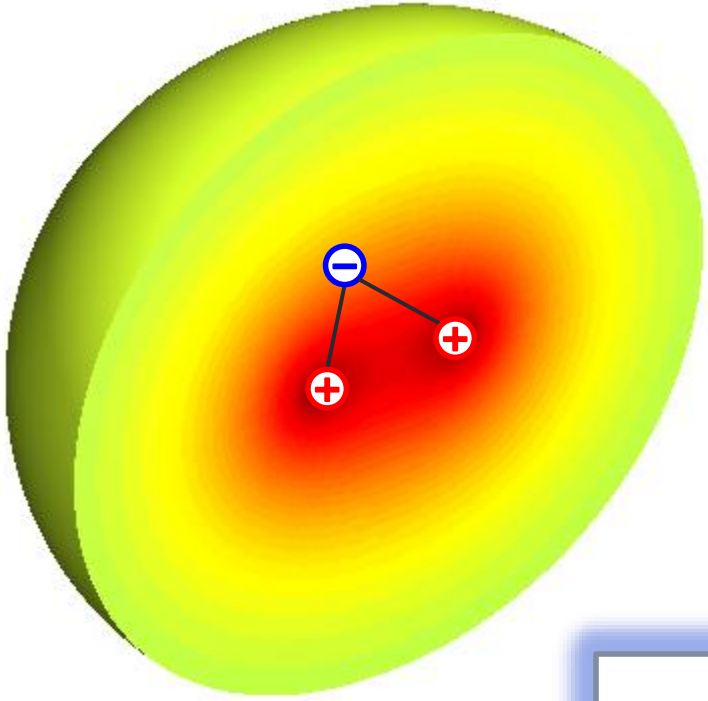
Molecular Orbital Theory

Exact solution available

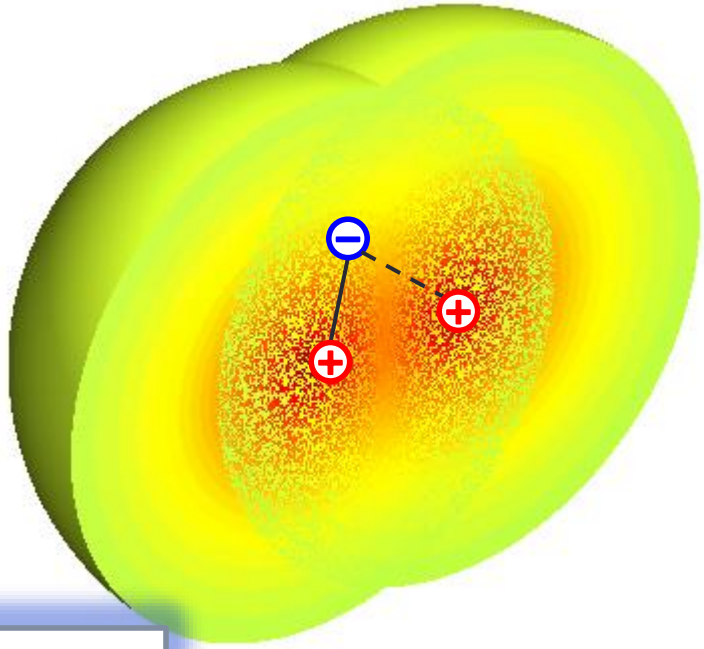


Each (valence) electron moves around all nuclei.

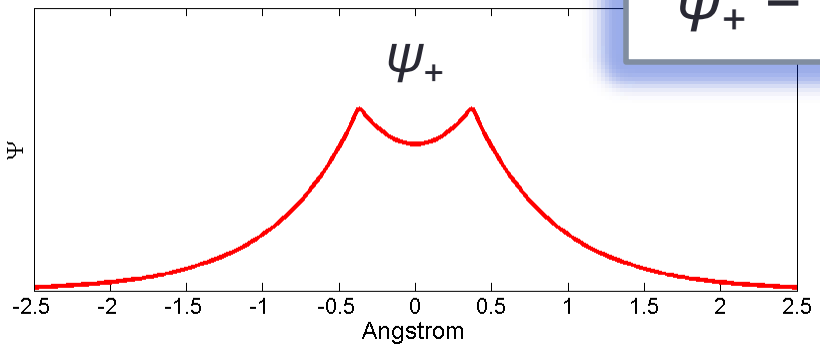
Linear Combination of Atomic Orbitals (LCAO)

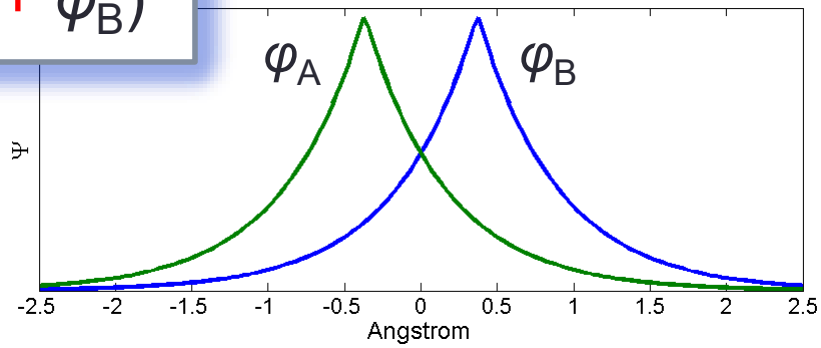


≈

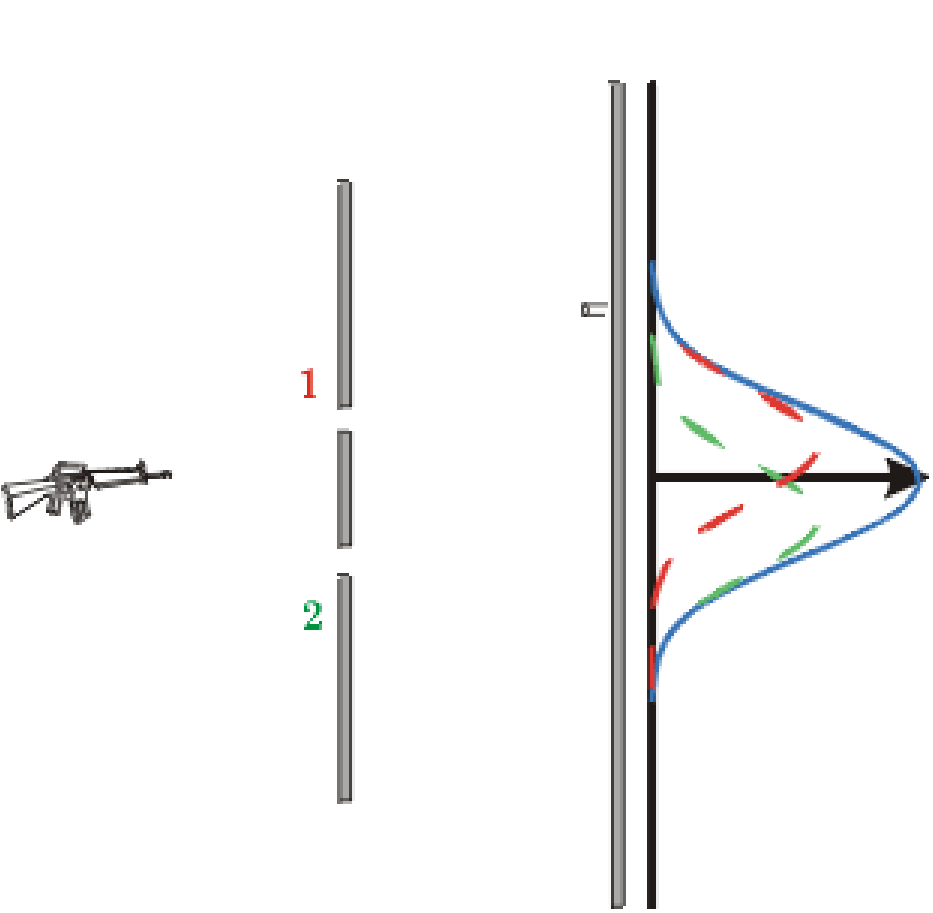


$$\psi_+ = c(\varphi_A + \varphi_B)$$

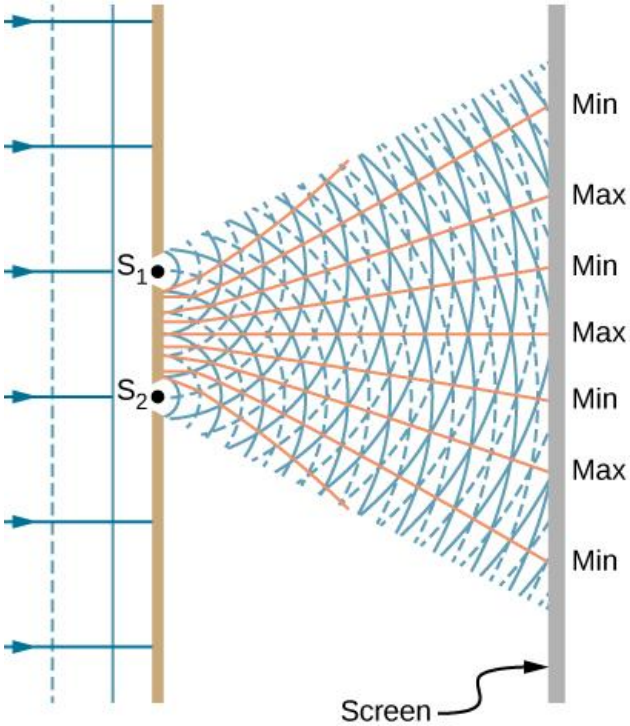




Double-Slit Experiment



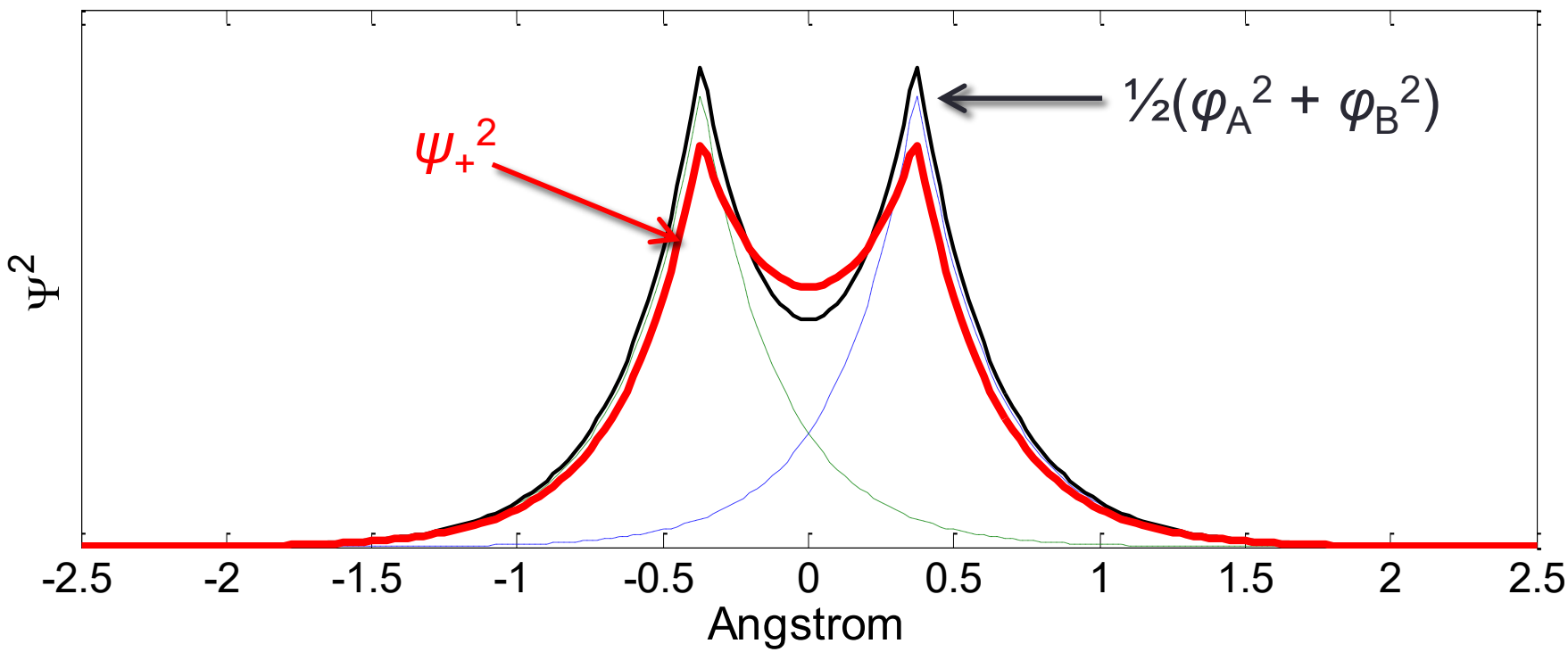
Mutually exclusive events
互斥事件



Wave interference

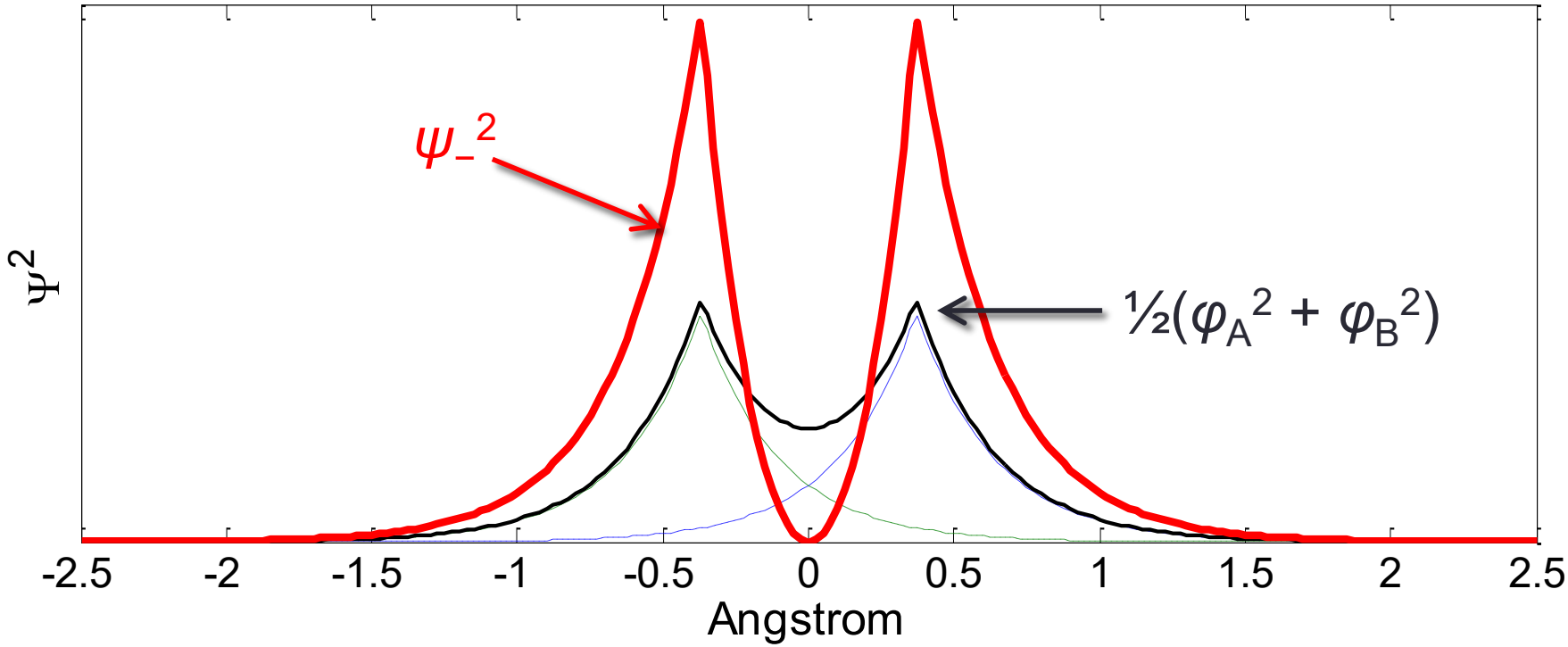
Bonding from Constructive Interference

$$\psi_+^2 = c^2(\varphi_A + \varphi_B)^2 = c^2(\varphi_A^2 + \varphi_B^2 + 2\varphi_A\varphi_B)$$

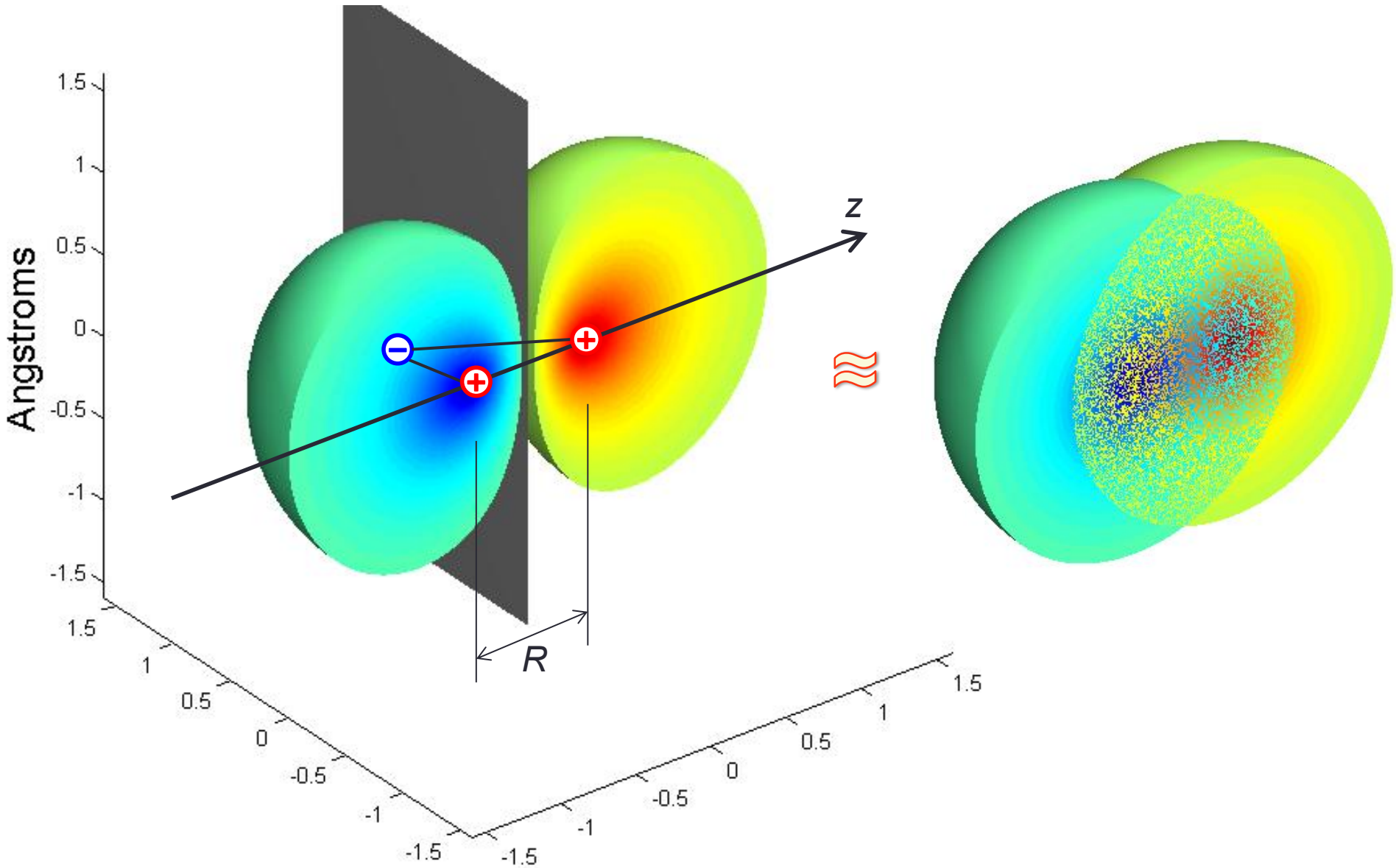


Antibonding from Destructive Interference
反键

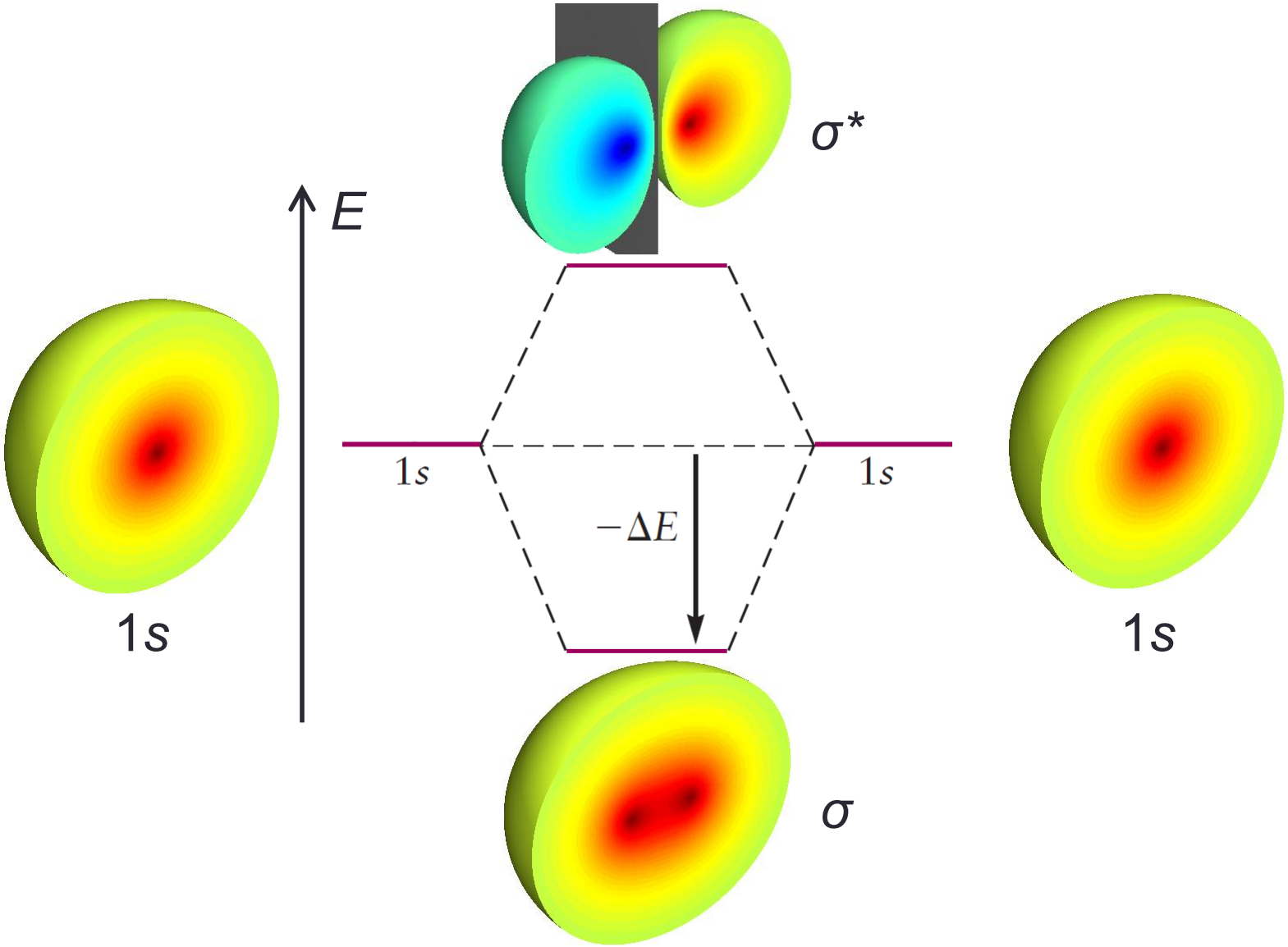
$$\psi_-^2 = c^2(\varphi_A - \varphi_B)^2 = c^2(\varphi_A^2 + \varphi_B^2 - 2\varphi_A\varphi_B)$$



Antibonding Orbital of H_2

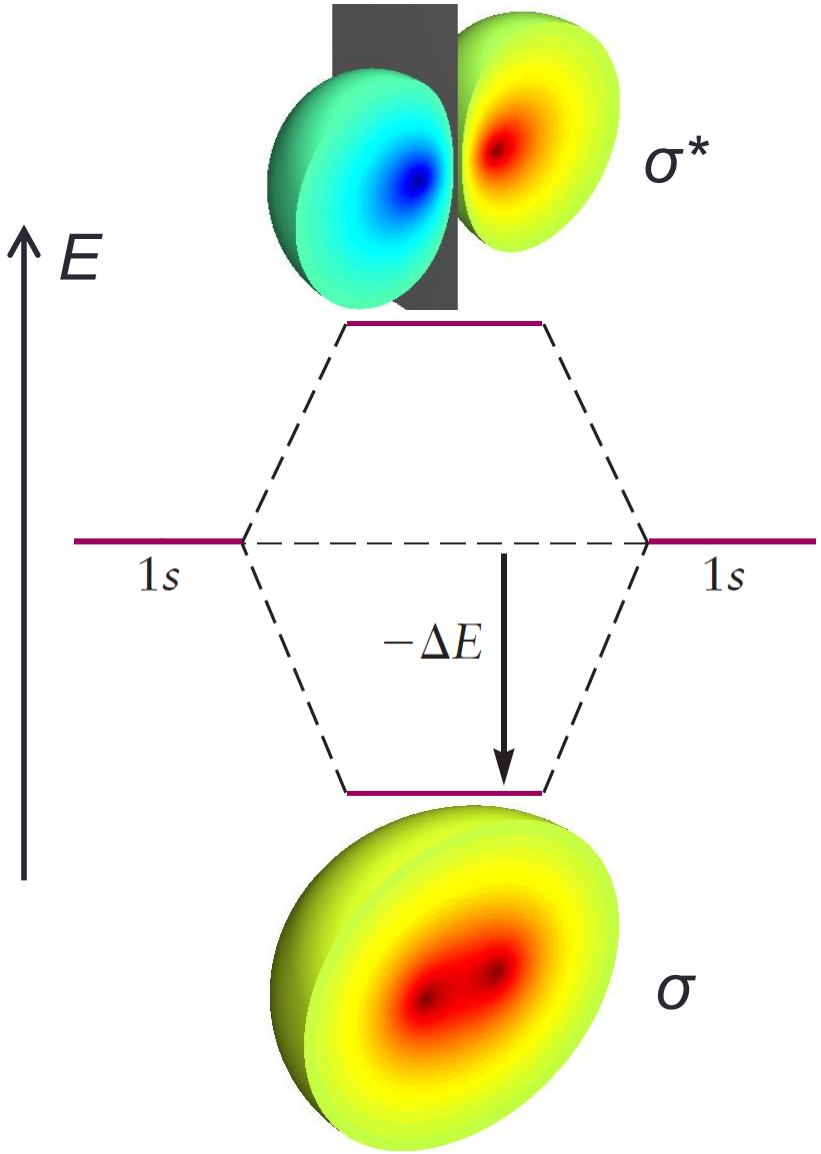


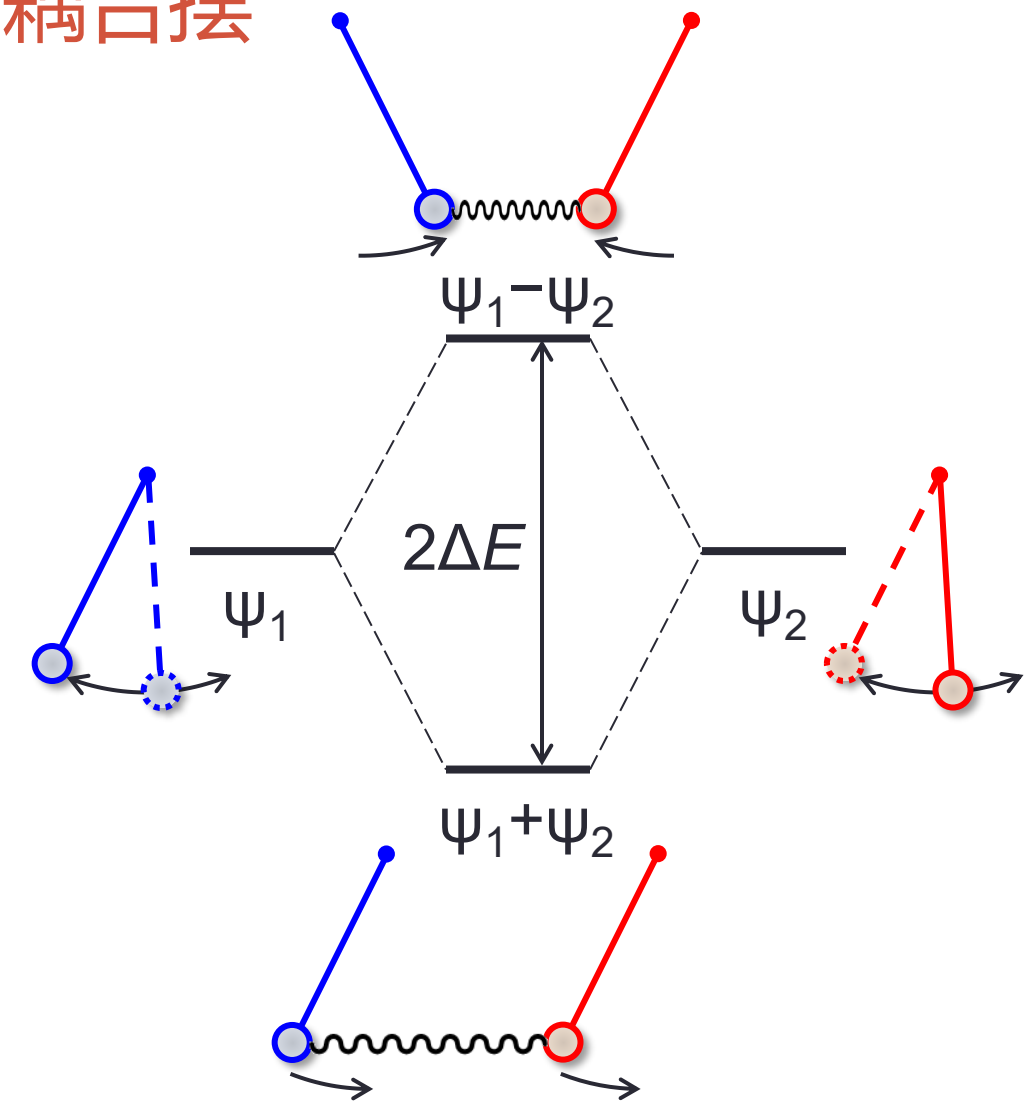
Orbital Correlation Diagram 轨道相关图



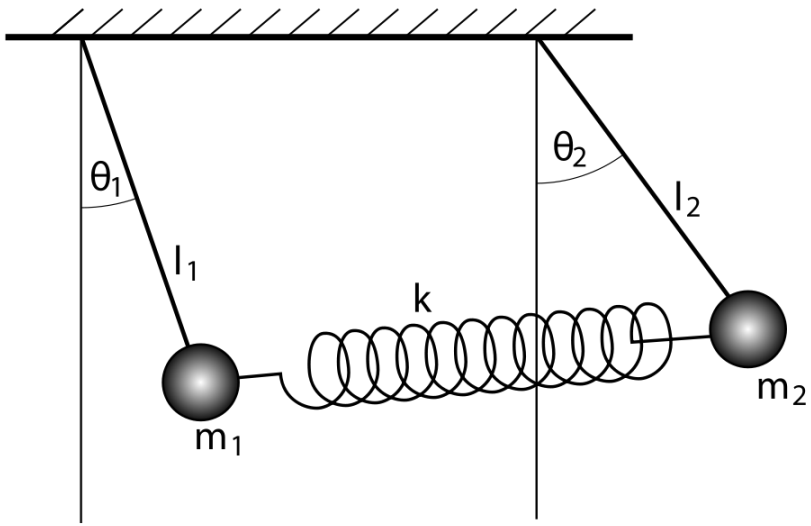
Antibonding and Coupled Pendula

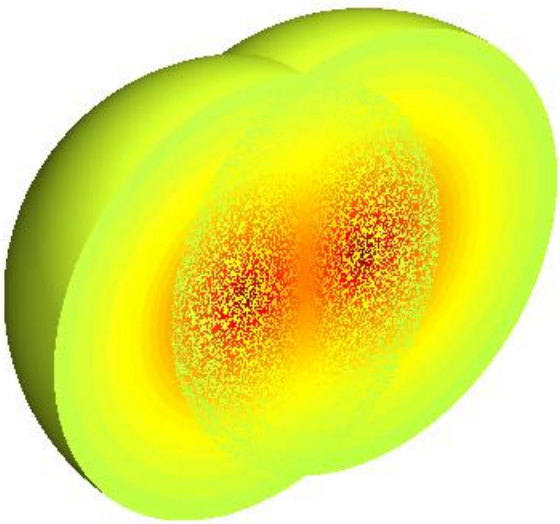
耦合摆





Coupled Pendula

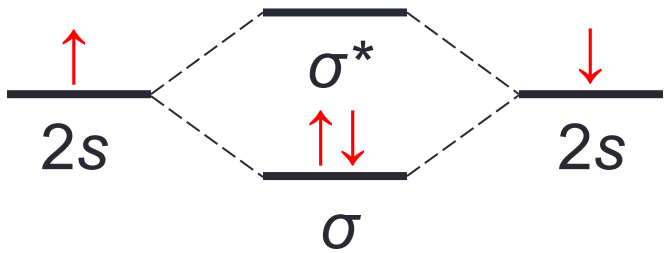


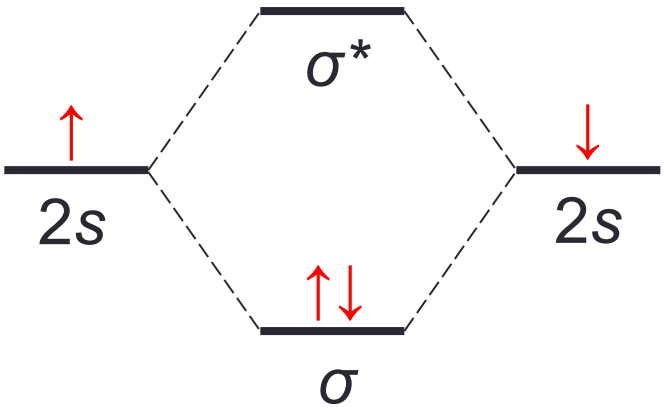


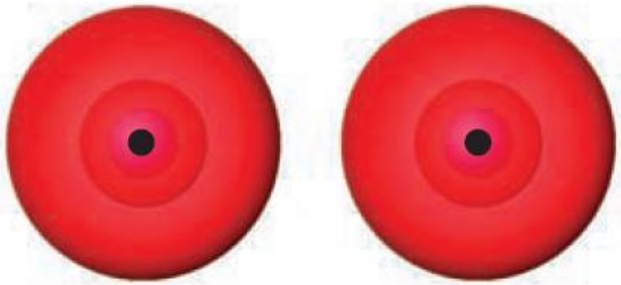
Coupled pendula		Molecular orbital
Angles θ_1, θ_2	$\Leftrightarrow$	Phases
Spring k	$\Leftrightarrow$	Orbital overlap
Masses m_1, m_2	$\Leftrightarrow$	Atomic orbital energies

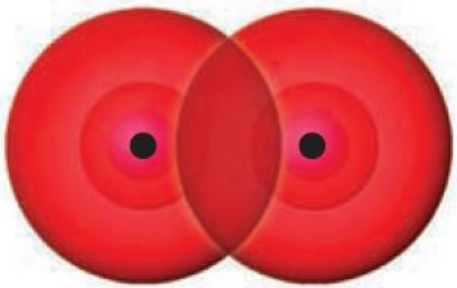
Factors for Effective Bonding

① Maximum orbital overlap



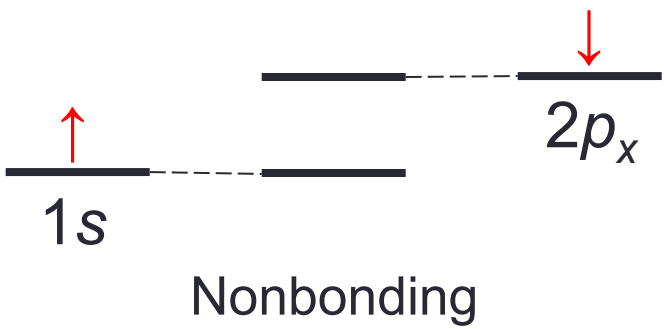


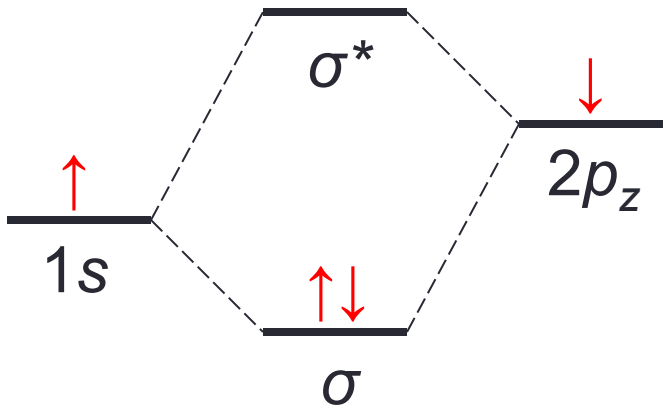


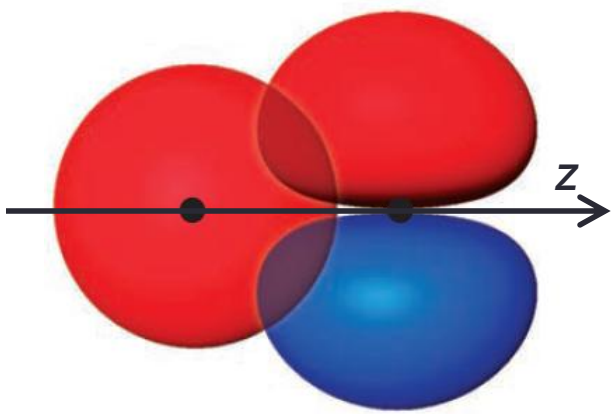


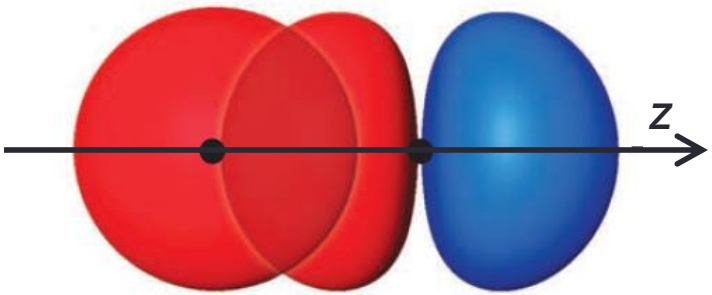
Factors for Effective Bonding

② Compatible orbital symmetries



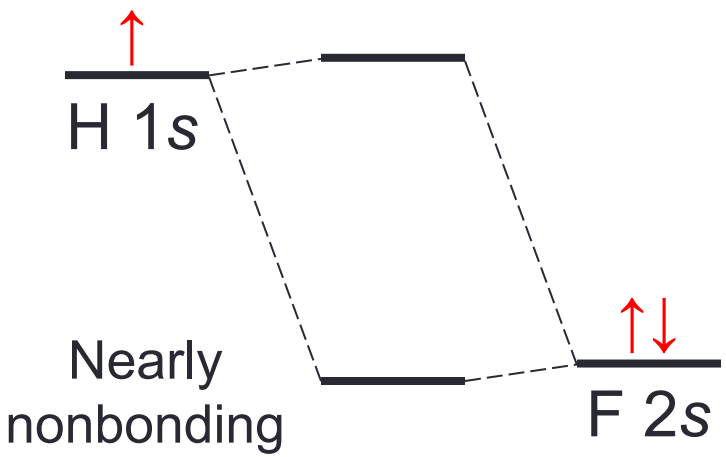


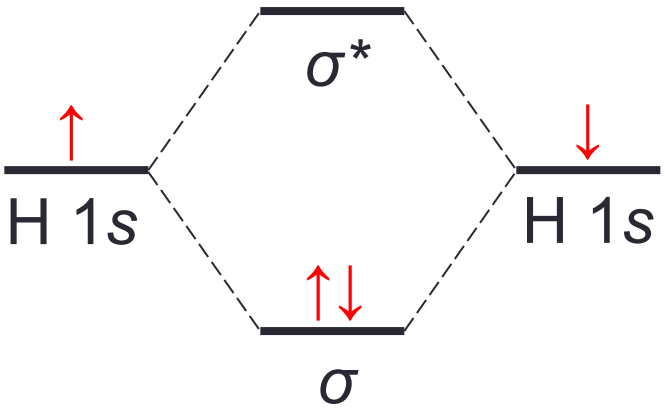


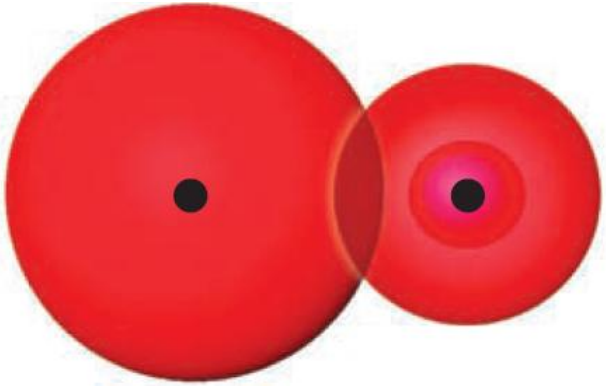


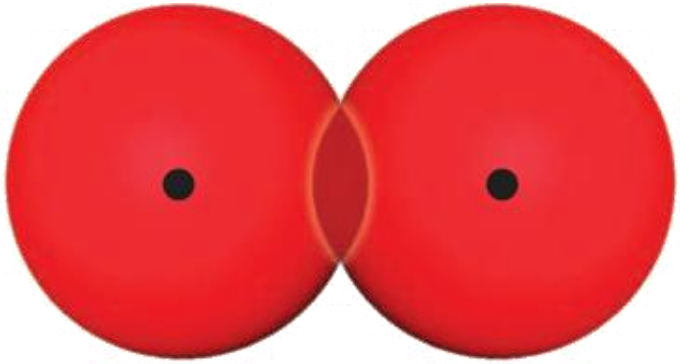
Factors for Effective Bonding

③ Approximate atomic orbital energies









Survey

Which of the three principles is the most effective?

A. Orbital overlap

B. Orbital symmetry

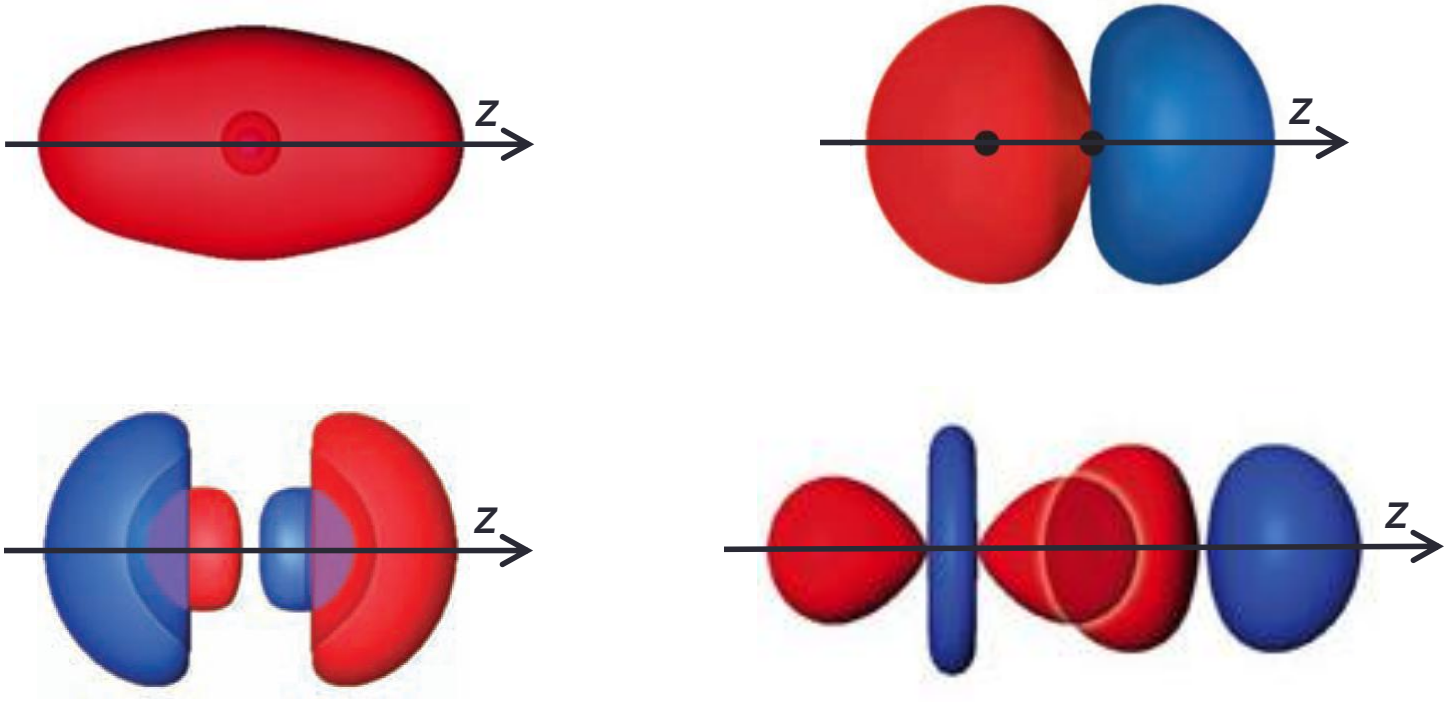
C. Orbital energy

Outline

- Overview
- Molecular Orbital Theory (1927)
- Types of bonding
  - σ, π bonds
  - Bonding, antibonding, nonbonding
  - Bond order
- Diatomic Molecules
- Valence Bond Theory (1927)
  - Hybridization (1930)

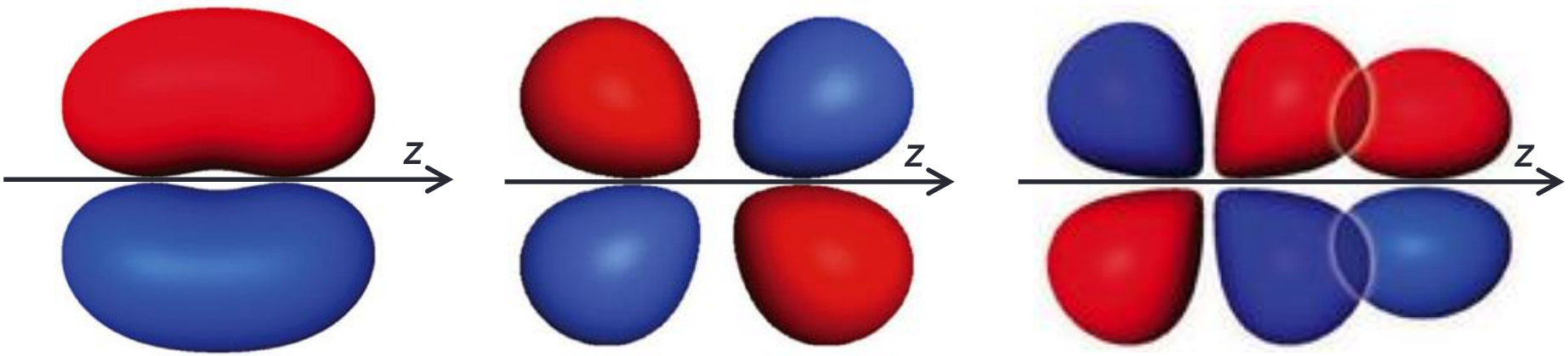
The σ Bond: $L_z = 0$

No nodal plane that contains the bond axis.



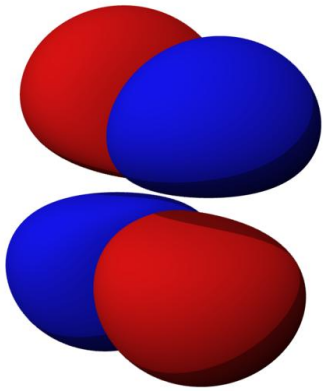
The π Bond: $L_z = \pm\hbar$

1 nodal plane that contains the bond axis.

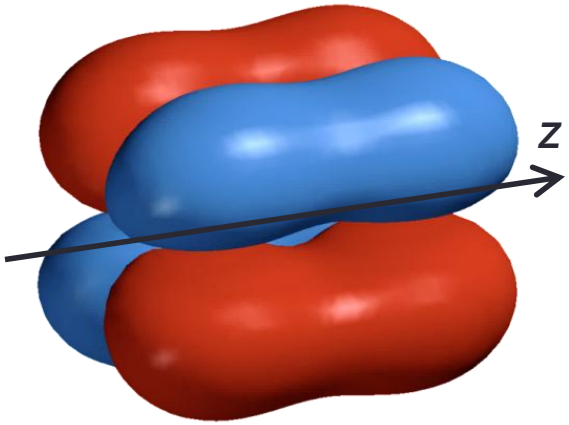


And more…

Nodal planes	Atomic orbital	L_z	Molecular orbital	L_z
0	*s*	0	σ	0
1	*p*	$-\hbar$, 0, $\hbar$	π	$\pm\hbar$
2	*d*	$-2\hbar$, $-\hbar$, 0, $\hbar$, $2\hbar$	δ	$\pm 2\hbar$



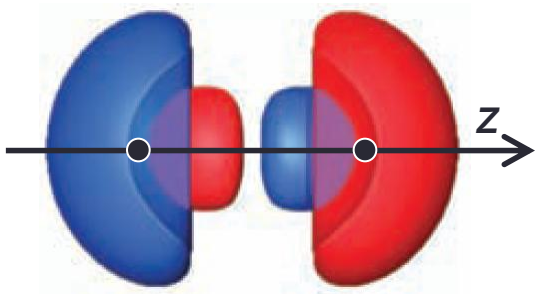
d orbital



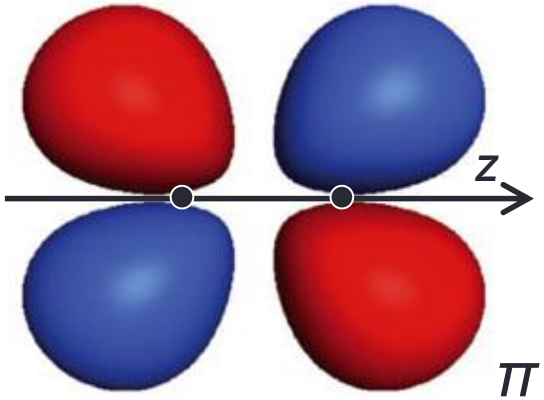
δ orbital

Node(s) in the Antibonding Orbital

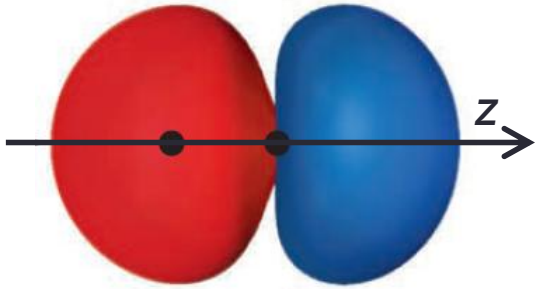
1 nodal plane in the middle of the bond.



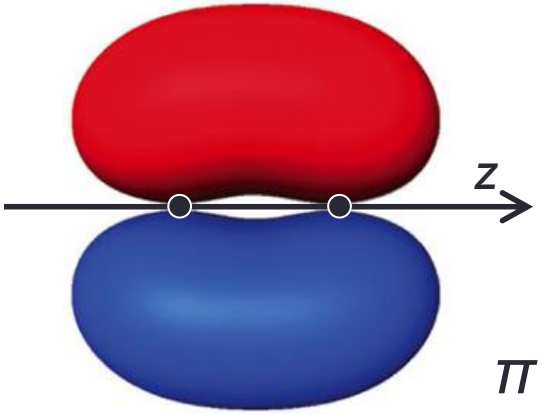
σ^*



π^*



σ



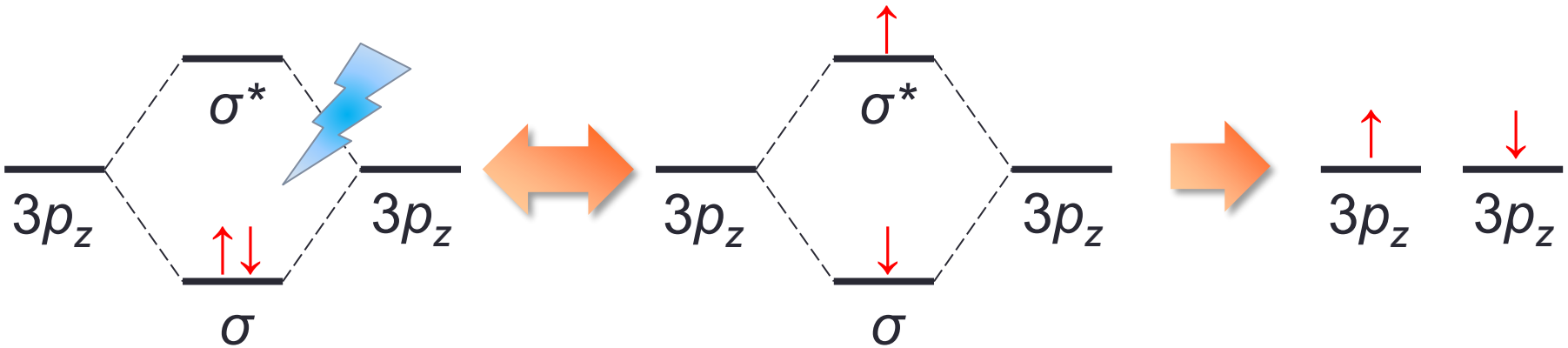
π

Meaning of the Antibonding Orbital

$H_2 + Cl_2 \longrightarrow 2\ HCl$

$$Cl{-}Cl \xrightarrow{h\nu} (Cl{-}{-}Cl)^*$$

$$\longrightarrow Cl^{\bullet} + Cl^{\bullet}$$

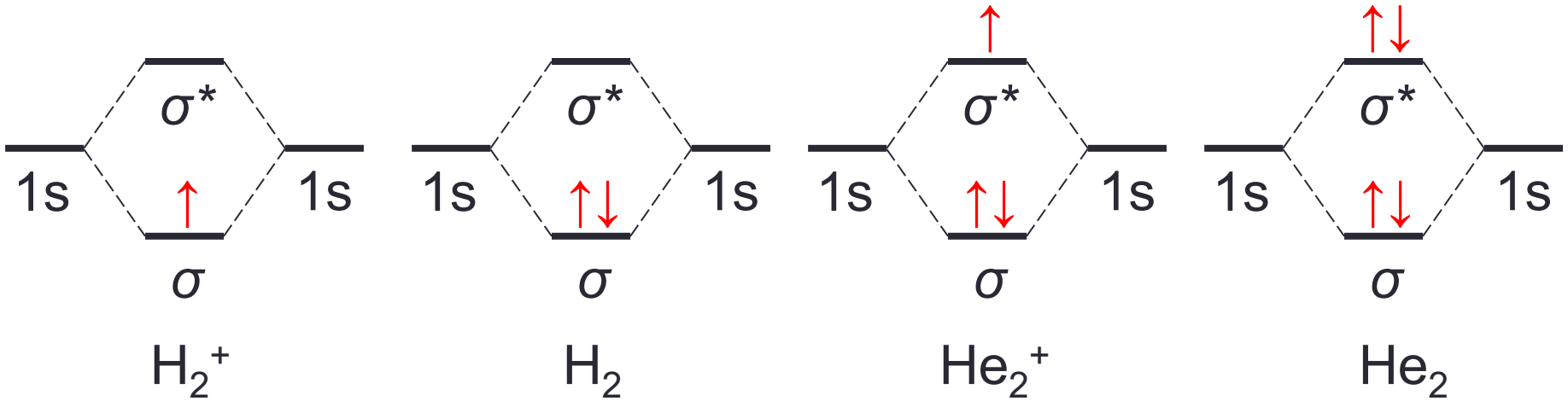


Bond Order 键级

Number of bonds		Bond order
Single	单键	1
Double	双键	2
Triple	叁键	3

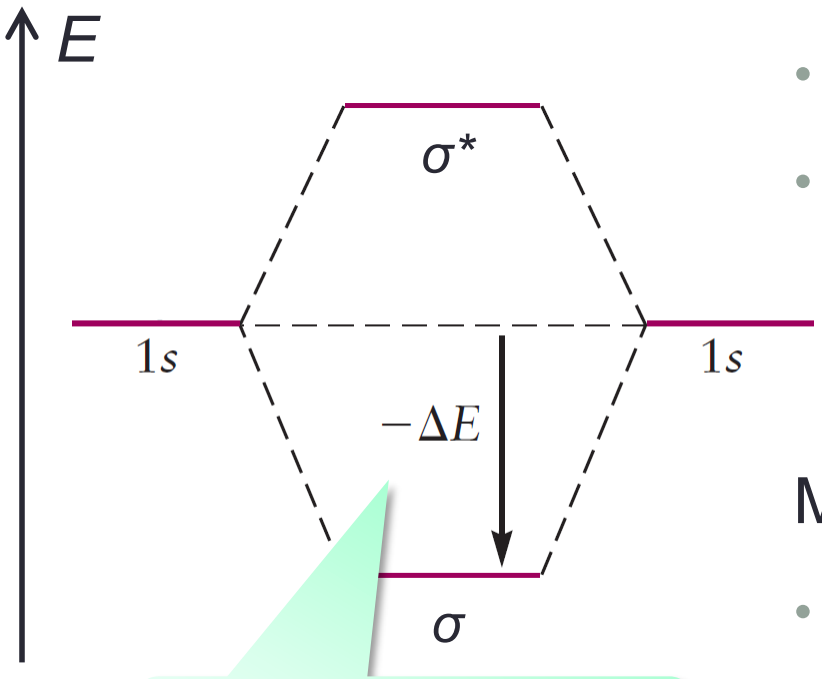
Bond order = No. of bonds − No. of antibonds =
(No. of bonding electrons − No. of antibonding electrons) ÷ 2

Bond Order (2)



Species	Electron Configuration	Bond Order	Bond Energy (kJ mol^{-1})	Bond Length (Å)
H_2^+	$(\sigma_{g1s})^1$	$\frac{1}{2}$	255	1.06
H_2	$(\sigma_{g1s})^2$	1	431	0.74
He_2^+	$(\sigma_{g1s})^2(\sigma^*_{u1s})^1$	$\frac{1}{2}$	251	1.08
He_2	$(\sigma_{g1s})^2(\sigma^*_{u1s})^2$	0	~0	Large

Features in the Correlation Diagram



How to estimate ΔE?

Molecular orbitals *do not* change:

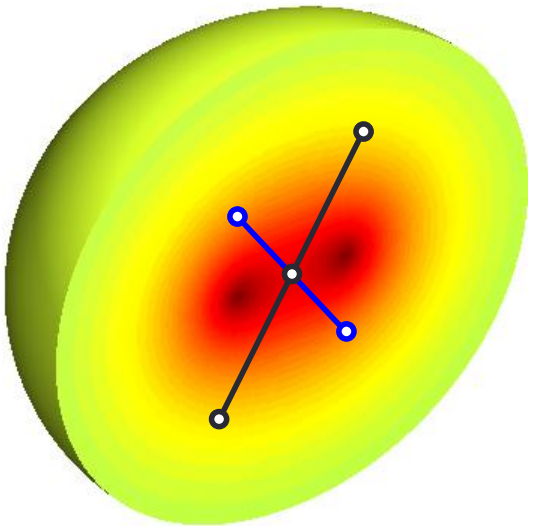
- Total number of orbitals
- Total energy of all orbitals
- Total number of electrons

Molecular orbitals *allow*

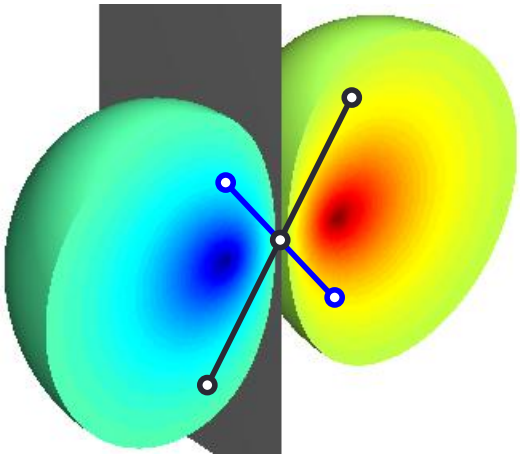
- Splitting of the orbital energies
- ≥ Half of the electrons to be more stable than the rest

Bond Symmetry

CN	EN	DE	Label
中心对称	Centrosymmetric	gerade	*g*
（中心）反对称	Antisymmetric	ungerade	*u*

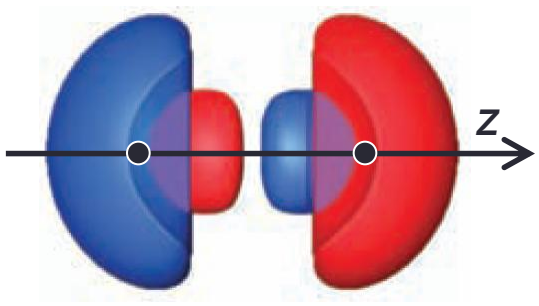


Centrosymmetric (*g*)

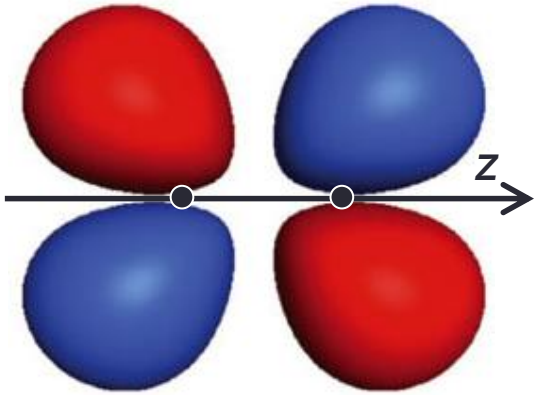


Antisymmetric (*u*)

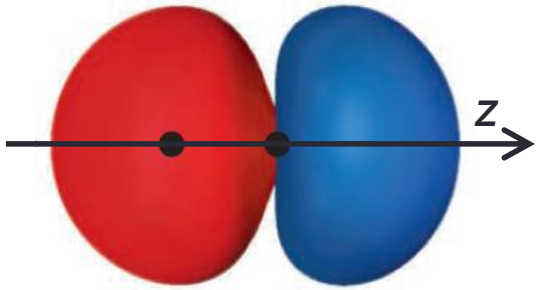
Combining Energy, Type & Symmetry



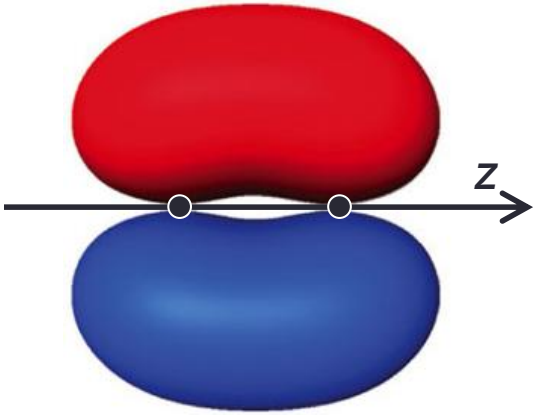
σ_u^*



π_g^*



σ



π_u

Outline

- Overview
- Molecular Orbital Theory (1927)
- Types of bonding
  - σ, π bonds
  - Bonding, antibonding, nonbonding
  - Bond order
- Diatomic Molecules
- Valence Bond Theory (1927)
  - Hybridization (1930)

Second-Period Elements

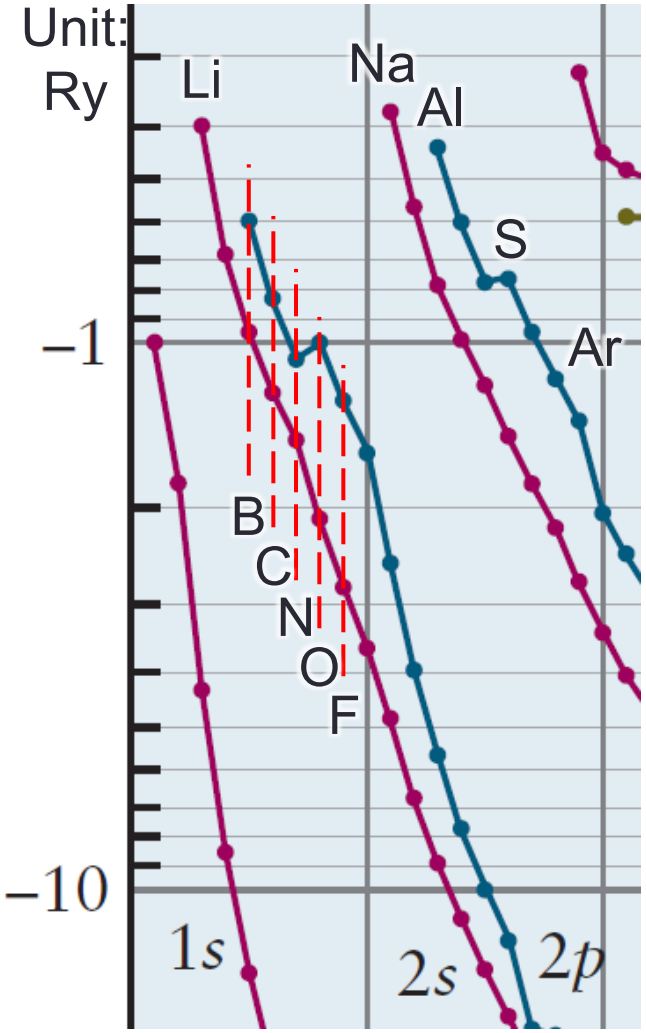
Orbital configuration and energy (in eV)

	B	C	N	O	F
2*s*	↑↓	↑↓	↑↓	↑↓	↑↓
	−14.1	−17.8	−25.4	−32.3	−40.2
2*p*	↑	↑ ↑	↑ ↑ ↑	↑↓ ↑ ↑	↑↓ ↑↓ ↑
	−8.3	−11.3	−14.5	−13.6	−17.4
2*p* − 2*s*	5.8	6.5	10.9	18.8	22.8

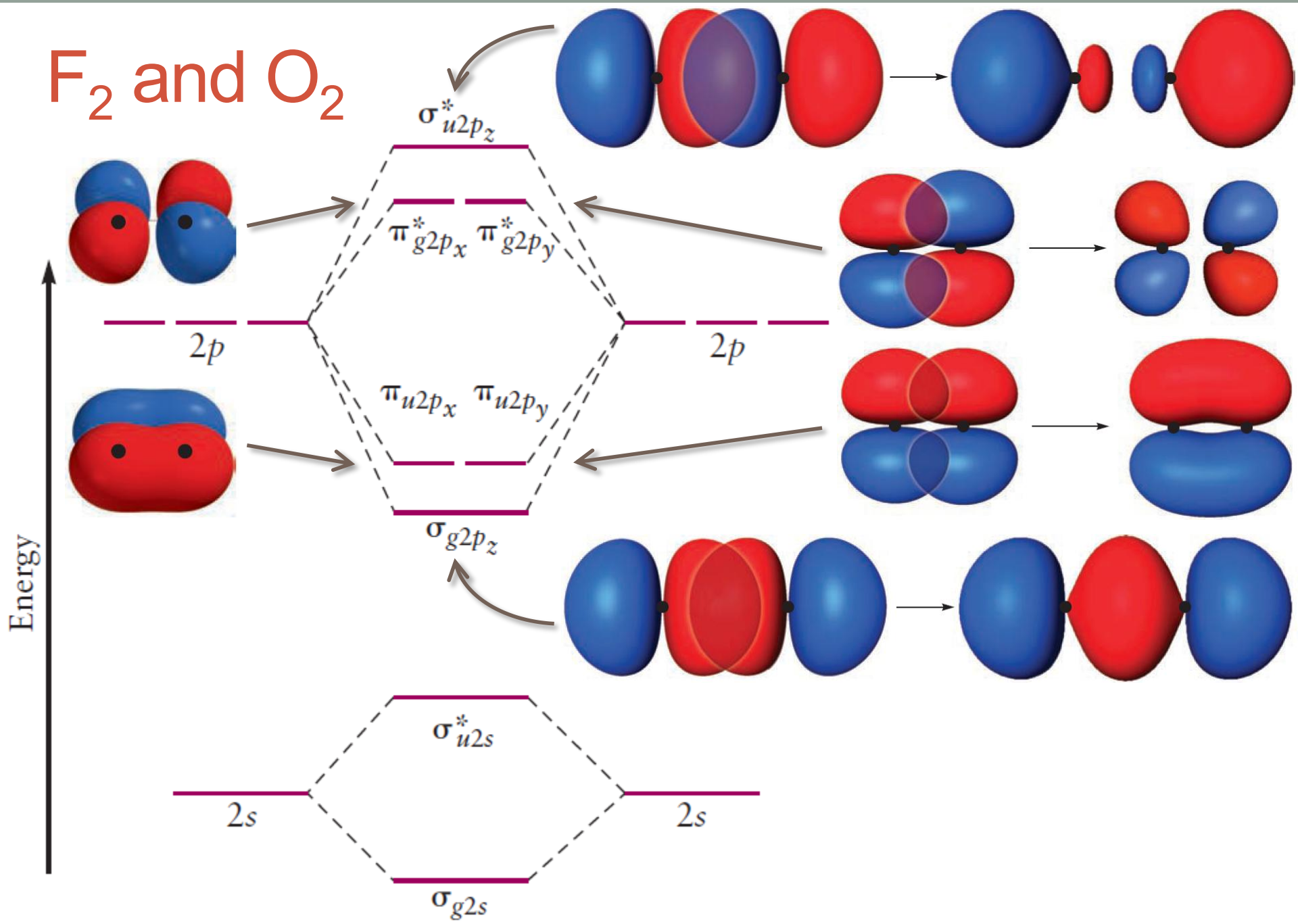
Small difference Large difference

The 2*s* and 2*p* orbitals of B, C, N tend to combine / hybridize 杂化 with each other.

Fig. 5.26, OGB8 p.210



F_2 and O_2


$\sigma^*_{u2p_z}$
$\pi^*_{g2p_x}$ $\pi^*_{g2p_y}$
2p
2p
π_{u2p_x} π_{u2p_y}
σ_{g2p_z}
Energy
σ^*_{u2s}
2s
2s
σ_{g2s}

Lewis, G. N. *J. Am. Chem. Soc.* **1924**, *46*, 2027–2032.

[CONTRIBUTION FROM THE CHEMICAL LABORATORY OF THE UNIVERSITY OF CALIFORNIA]

THE MAGNETISM OF OXYGEN AND THE MOLECULE O_4

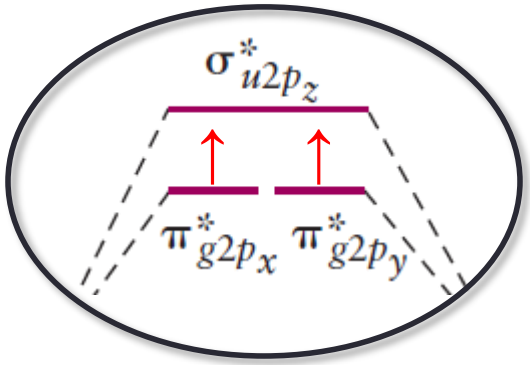
BY GILBERT N. LEWIS

RECEIVED JUNE 25, 1924 PUBLISHED SEPTEMBER 5, 1924

One of the most characteristic properties of oxygen is its pronounced paramagnetism. This distinguishes molecular oxygen so sharply from other compounds in which oxygen is assumed to be tied by a double bond that I have recently concluded[1] that there is no double bond in the O_2 molecule. Rather I have assumed that the two electrons which have been supposed to constitute the second bond are actually separate so that each atom has an odd electron, according to the formula $:\ddot{\underset{\cdot}{O}}:\ddot{\underset{\cdot}{O}}:$. Indeed, oxygen has the chemical properties of a molecule with odd electrons, for it has an appreciable color, and a much higher reactivity than has commonly been ascribed to it.

O_2: Unusual Properties (1)

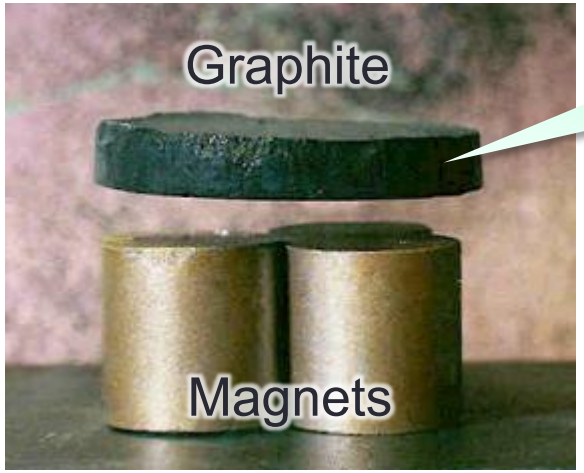
O_2 is paramagnetic due to its two unpaired electron spins.



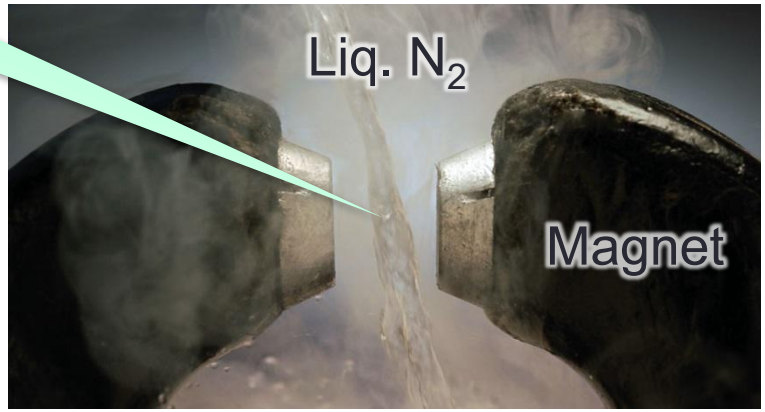
Paramagnetic 顺磁性

Spinning electrons

Orbiting electrons



Diamagnetic 反（抗、逆）磁性

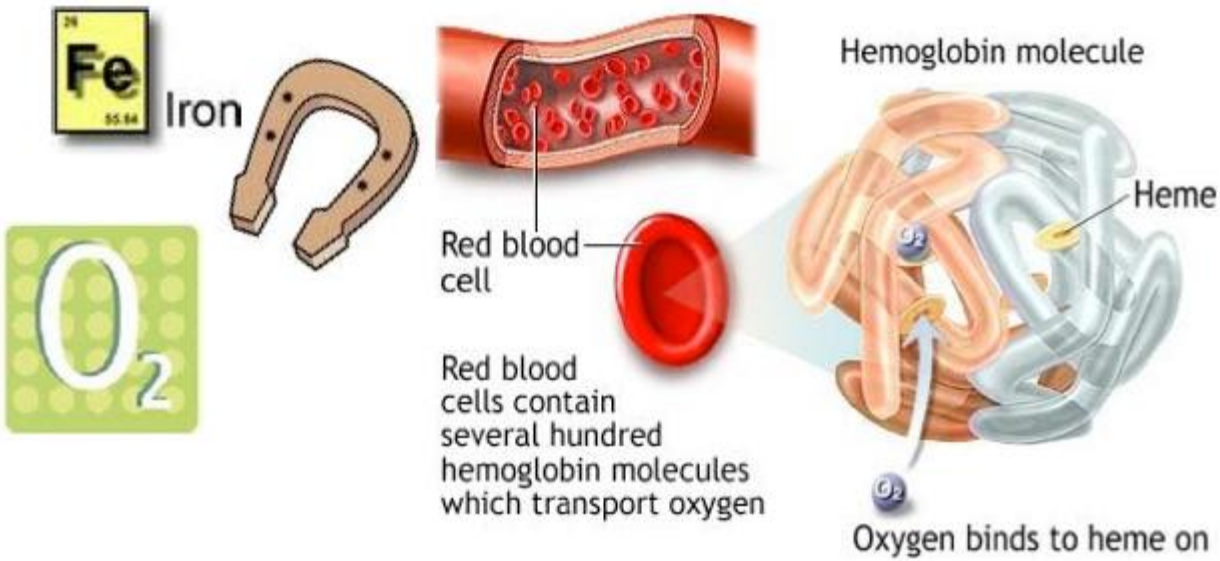


Weakly diamagnetic

O_2: Unusual Properties (2)

- Liquid O_2 is pale blue.

- O_2 reacts readily with transition metals, but slowly with common organics at r.t.



$$H_2C{=}CH_2 + Br_2 \xrightarrow[\text{r.t.}]{\text{Fast}} CH_2BrCH_2Br$$

$$H_2C{=}CH_2 + O_2 \xrightarrow[\text{r.t.}]{} \times$$

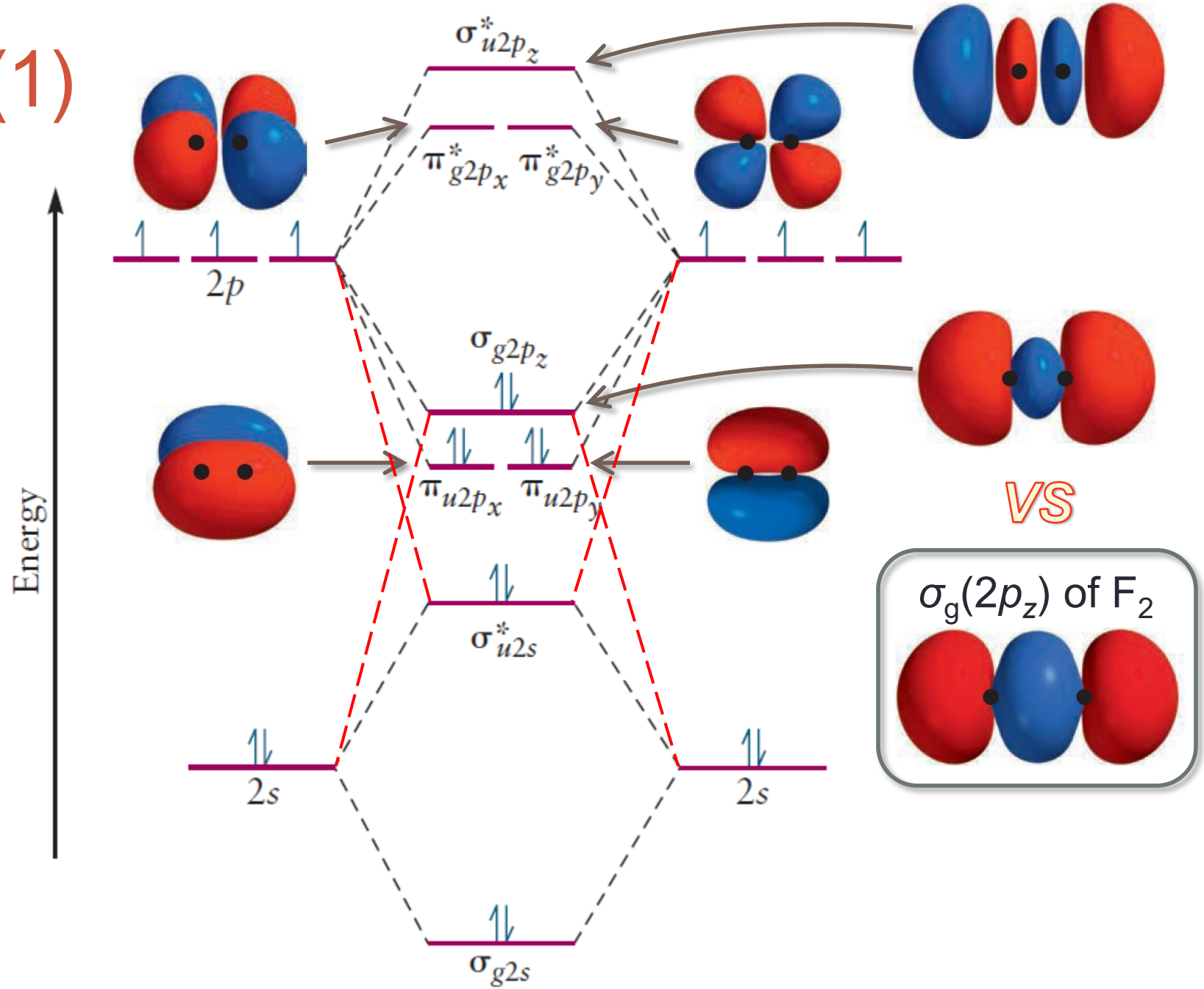
Other Species Related to O_2

	F_2	O_2^{2-}	O_2^-	O_2	O_2^+
Electronic configuration	$\sigma^2\pi^4\pi^{*4}$	$\sigma^2\pi^4\pi^{*4}$	$\sigma^2\pi^4\pi^{*3}$	$\sigma^2\pi^4\pi^{*2}$	$\sigma^2\pi^4\pi^{*1}$
Bond order	1	1	1.5	2	2.5
Bond length (Å)	1.41	1.49	1.33	1.21	1.12
Bond energy (kJ/mol)	154	142	~370	494	625

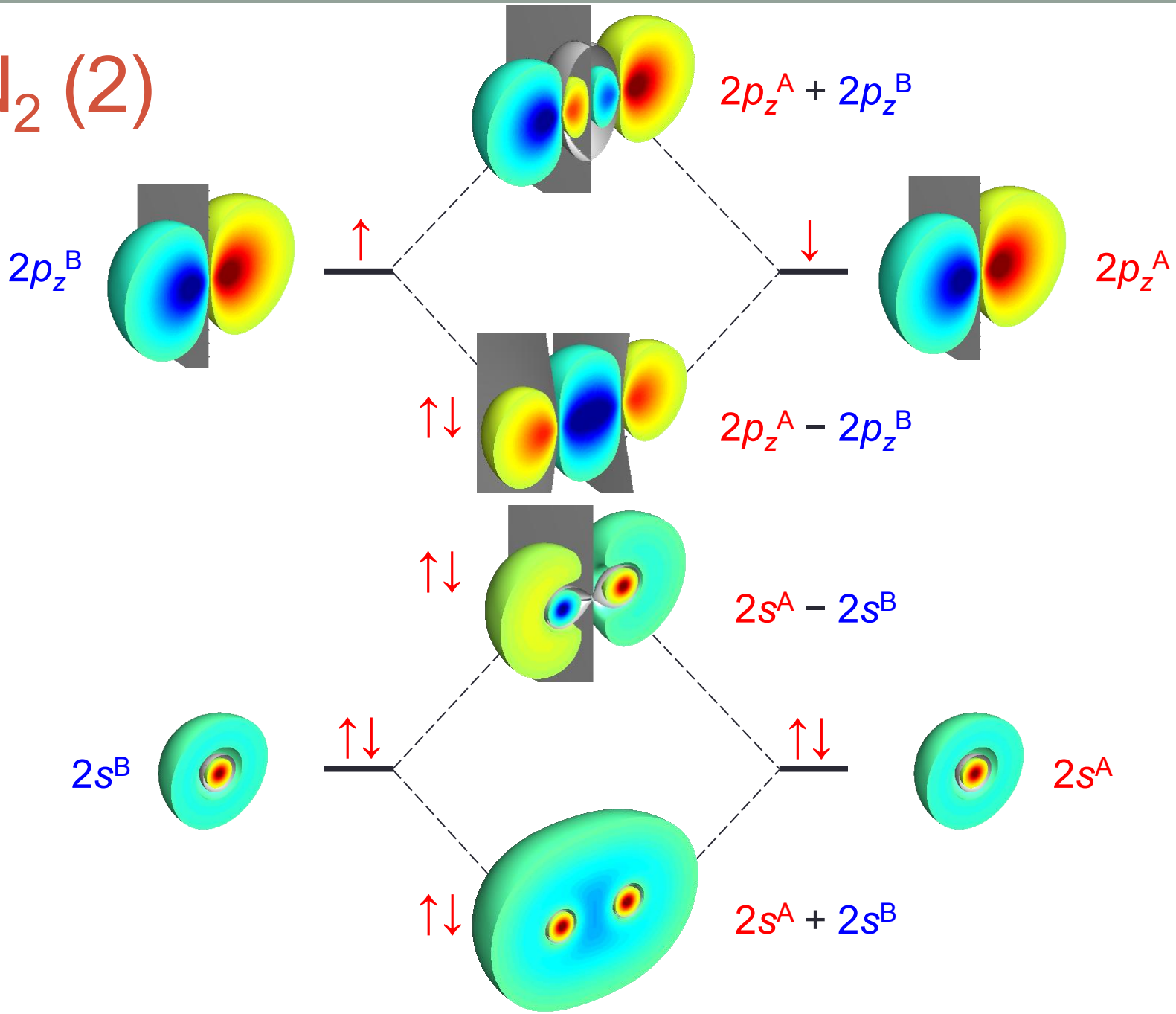
Isoelectronic

$$O_2 + PtF_6 \longrightarrow O_2^+[PtF_6]^- \quad (1962)$$

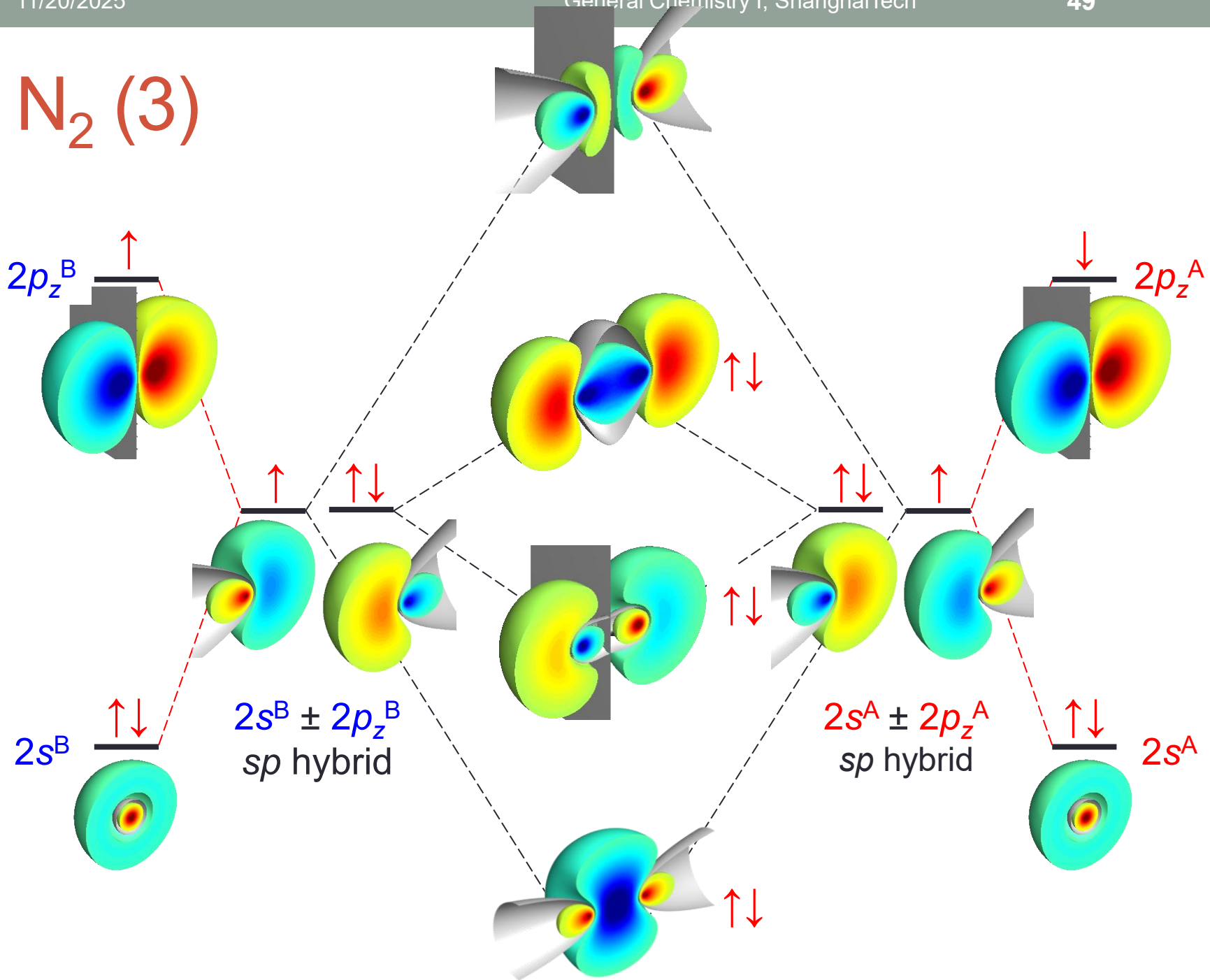
N_2 (1)


$\sigma^*_{u2p_z}$
$\pi^*_{g2p_x}$ $\pi^*_{g2p_y}$
2p
σ_{g2p_z}
π_{u2p_x} π_{u2p_y}
σ^*_{u2s}
2s
2s
σ_{g2s}
Energy
VS
$\sigma_g(2p_z)$ of F_2

N_2 (2)



N_2 (3)



Other Species Related to N_2

	O_2	O_2^+	NO	NO^+	N_2	CO	CN^-	C_2^{2-}
Electronic configuration	$\sigma^2\pi^4\pi^{*2}$	$\sigma^2\pi^4\pi^{*1}$	$\pi^4\sigma^2\pi^{*1}$	$\pi^4\sigma^2$	$\pi^4\sigma^2$	$\pi^4\sigma^2$	$\pi^4\sigma^2$	$\pi^4\sigma^2$
Bond order	2	2.5	2.5	3	3	3	3	3
Bond length (Å)	1.21	1.12	1.15	1.06	1.10	1.13	1.15	1.20
Bond energy (kJ/mol)	494	625	607	?	942	1070	891	839

Isoelectronic

Isoelectronic

$$CaO + 3\,C \xrightarrow{2000\ ^\circ C} CaC_2 + CO$$

LCAO-MO: Summary

福井谦一
(Kyoto, 1918–1998)

Atomic orbitals and molecular orbitals

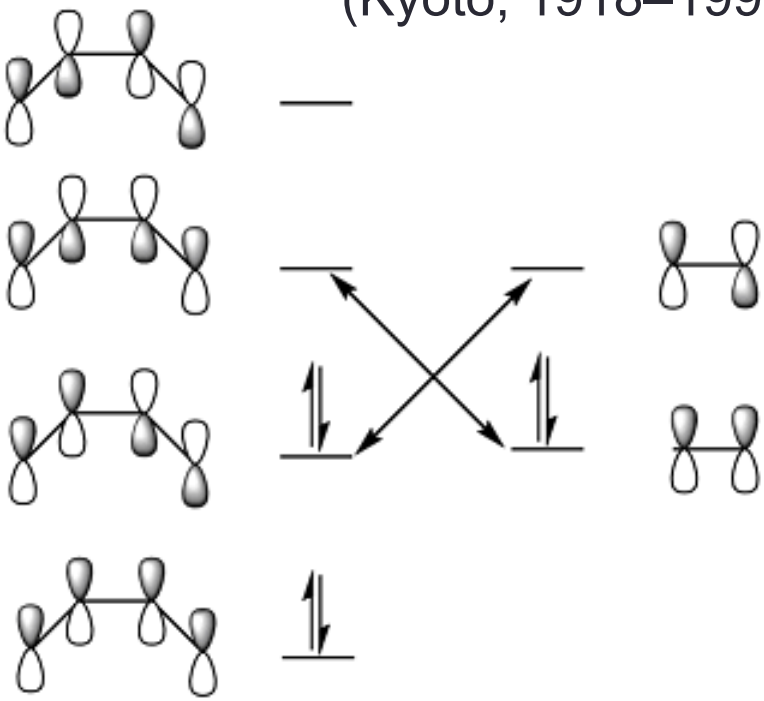
- Can be linearly combined
- Obey the Aufbau Principle

Rules of bonding

- Maximum orbital overlap
- Compatible orbital symmetries
- Approximate atomic orbital energies
- No. of electrons ≈ No. of orbitals

Frontier orbitals (1952)

前线/前沿轨道



Outline

- Overview
- Molecular Orbital Theory (1927)
- Types of bonding
  - σ, π bonds
  - Bonding, antibonding, nonbonding
  - Bond order
- Diatomic Molecules
- Valence Bond Theory (1927)
  - Hybridization (1930)

MO *vs.* VB Revisited

Molecular Orbital (MO)

- Ignores e-e repulsion
- Electrons wander around
- For orbitals that are
  - Delocalized 离域
  - 'Fat'
  - Singly occupied

Valence Bond (VB)

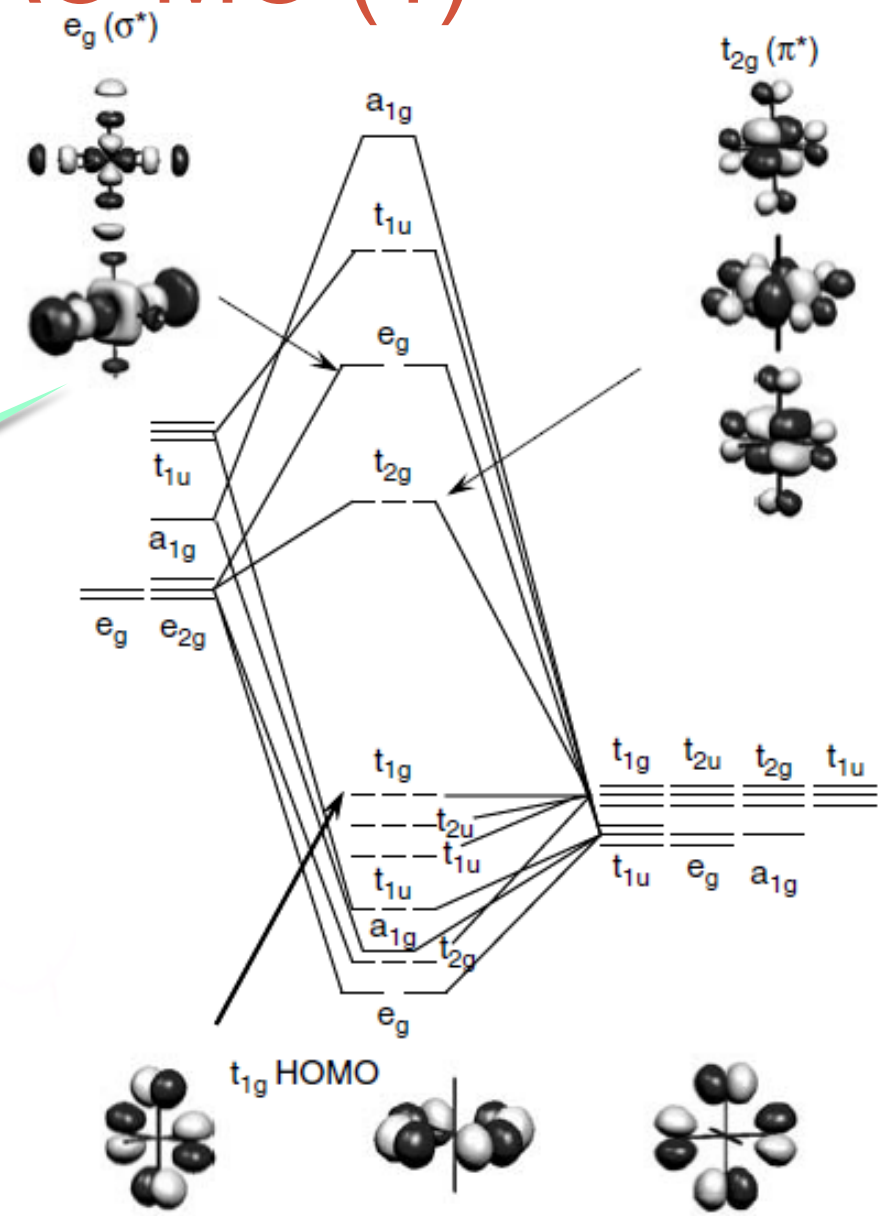
- Exaggerates e-e repulsion
- Electrons adhere to the nuclei
- For orbitals that are
  - Localized 定域/局域
  - 'Tight'
  - Strong

Weaknesses of LCAO-MO (1)

Many atomic orbitals combine to make one molecular orbital.

- Non-intuitive
- Require computer codes

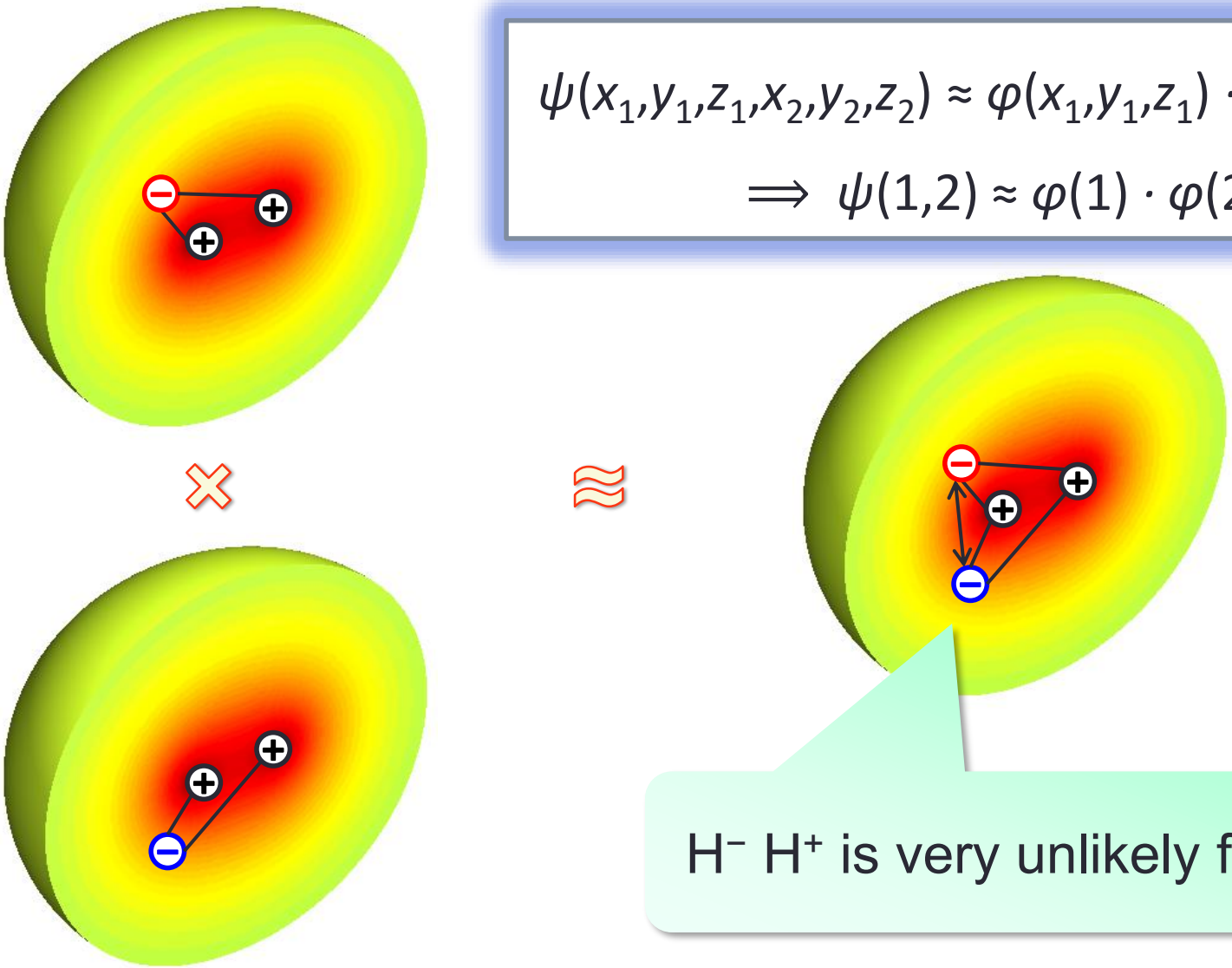
TiO_6^{8-}
(in crystal)



Weaknesses of LCAO-MO (2)

$$\psi(x_1,y_1,z_1,x_2,y_2,z_2) \approx \varphi(x_1,y_1,z_1) \cdot \varphi(x_2,y_2,z_2)$$

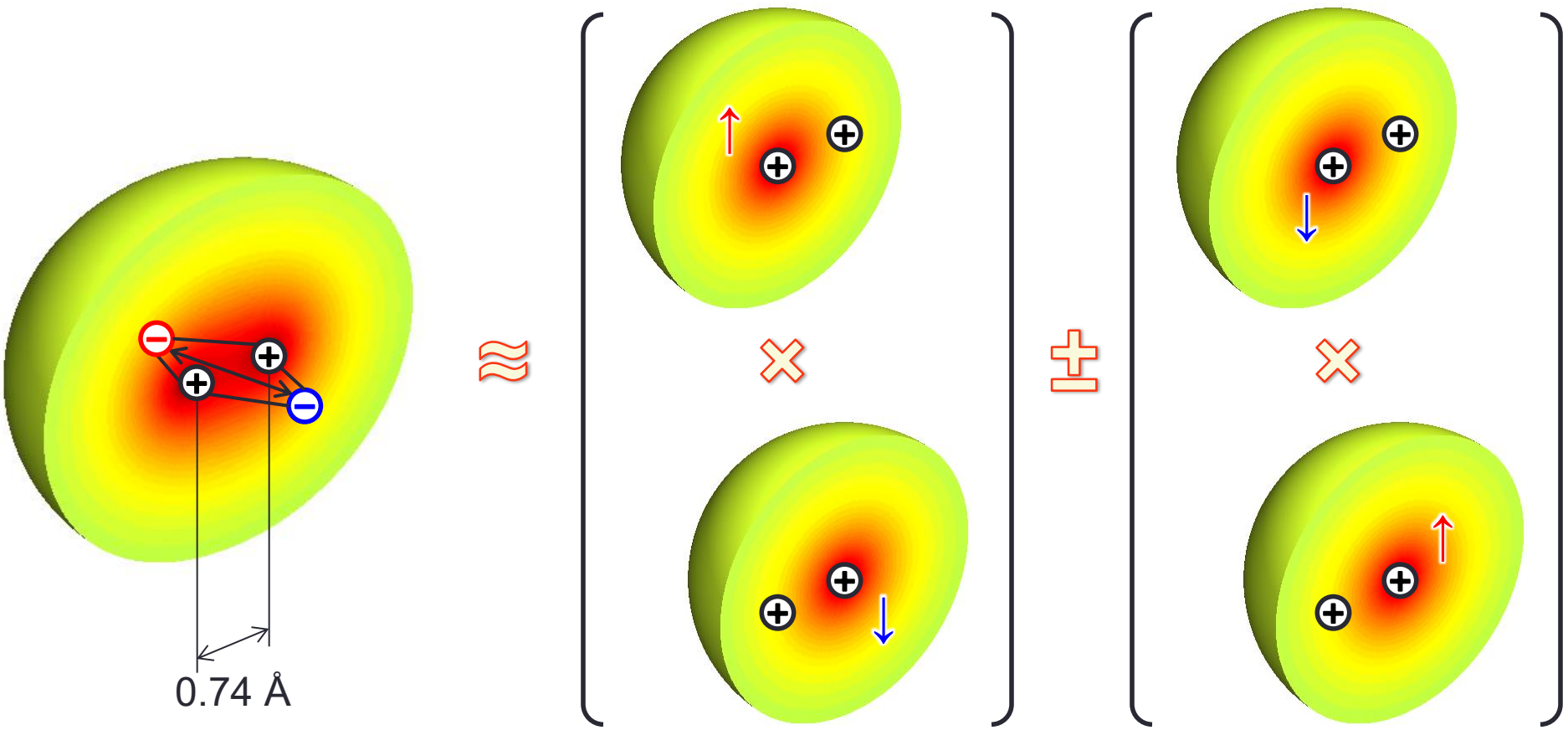
$$\Longrightarrow \psi(1,2) \approx \varphi(1) \cdot \varphi(2)$$



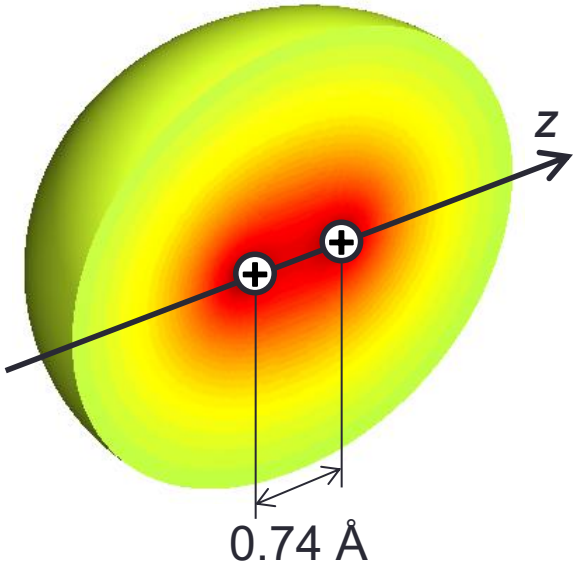
H^- H^+ is very unlikely for H_2!

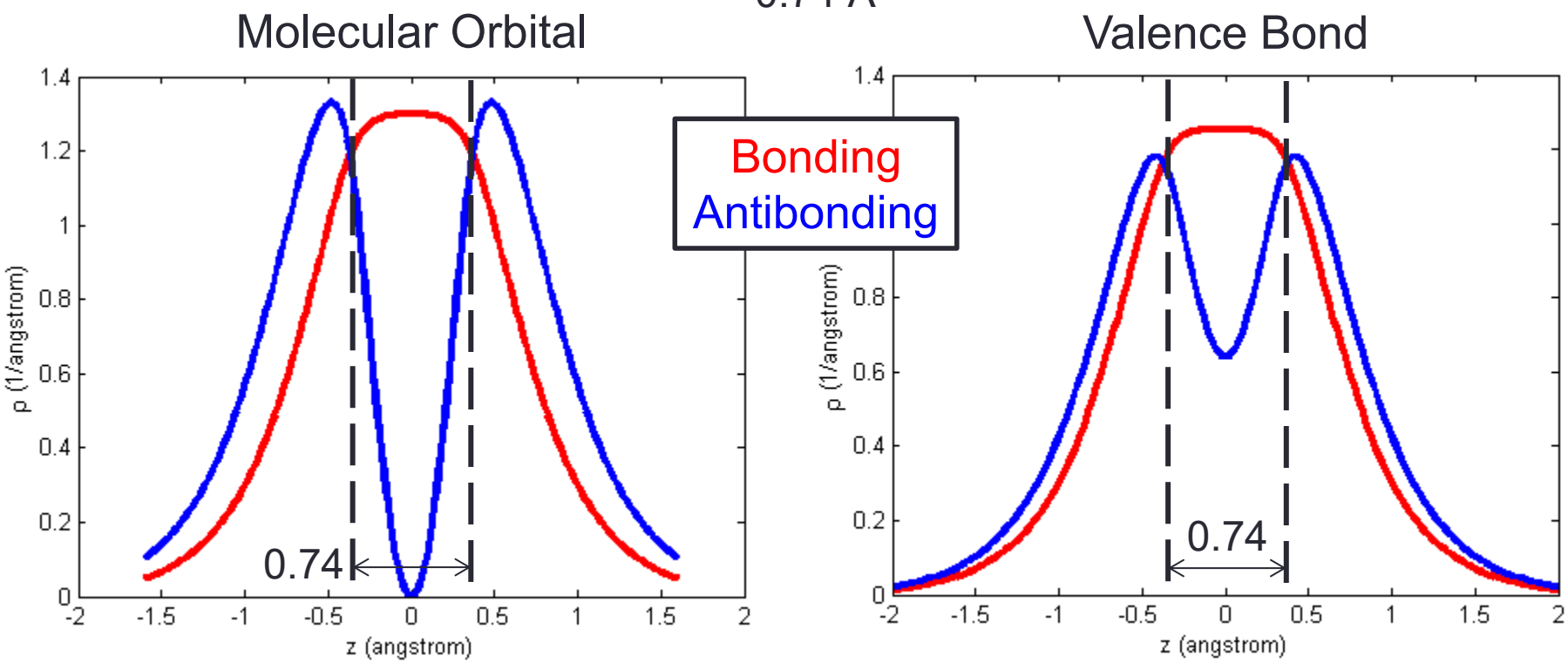
The Valence Bond Theory

$$\psi_{\pm}(x_1,x_2) \approx c_1[\varphi^{A}(x_1)\bullet\varphi^{B}(x_2) \pm \varphi^{A}(x_2)\bullet\varphi^{B}(x_1)]$$



Comparison between MO & VB





Pros and Cons of VB Theory

Works best for localized bonds

Difficult to estimate energy levels

Complicated programming

More accurate than MO for σ bonds

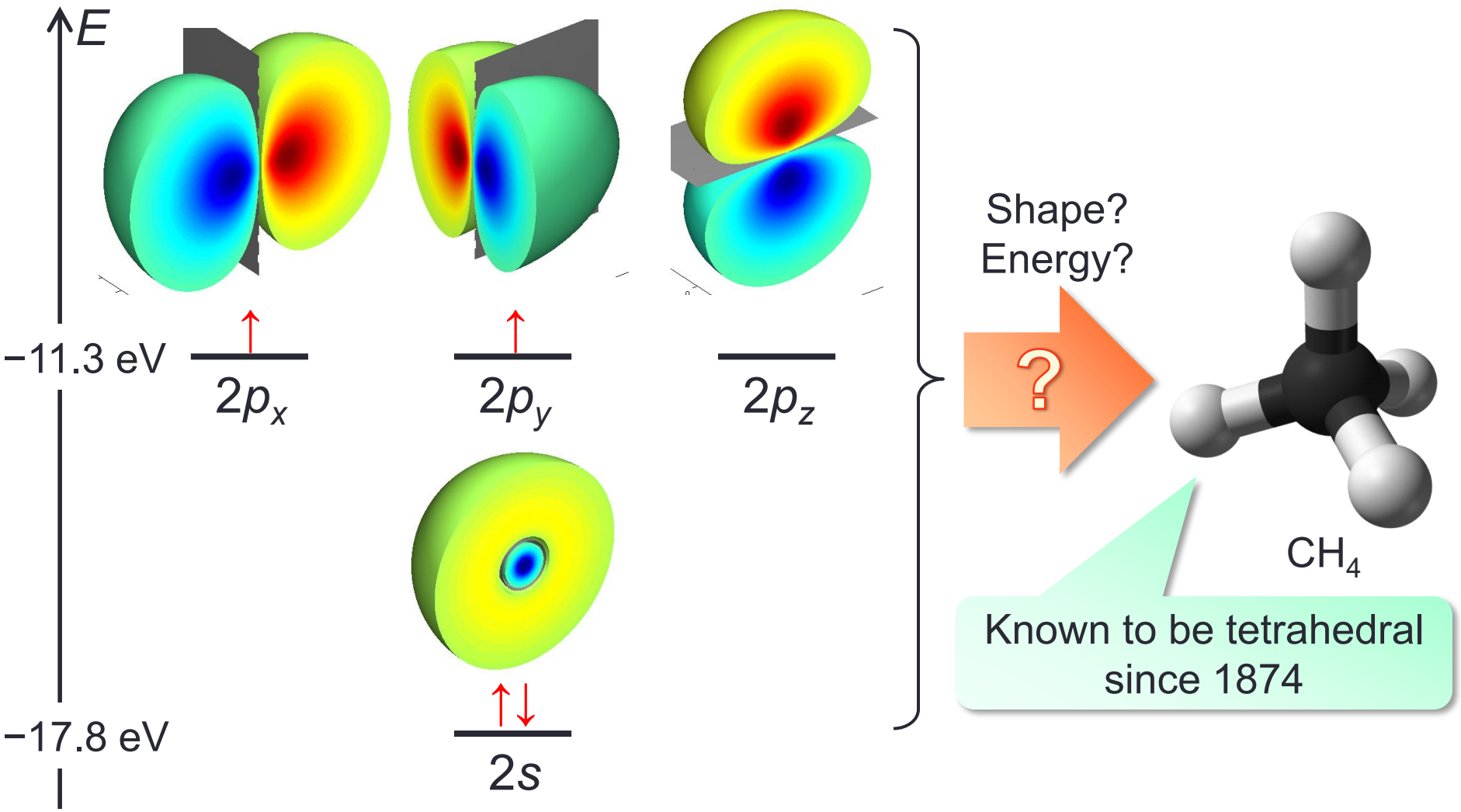
Compatible with hybridization 杂化

$$\psi_g(x_1,x_2) \approx c_1[\varphi^A(x_1)\bullet\varphi^B(x_2) + \varphi^A(x_2)\bullet\varphi^B(x_1)]$$

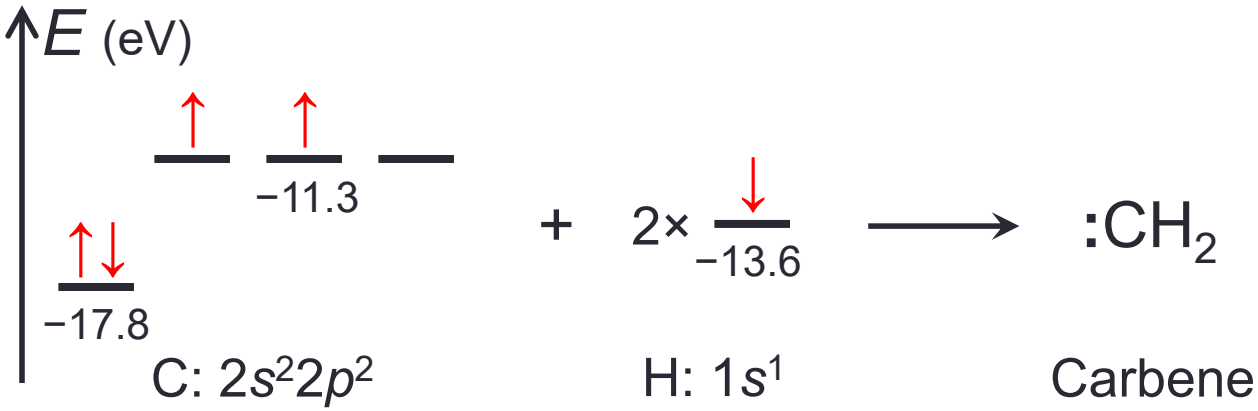
Outline

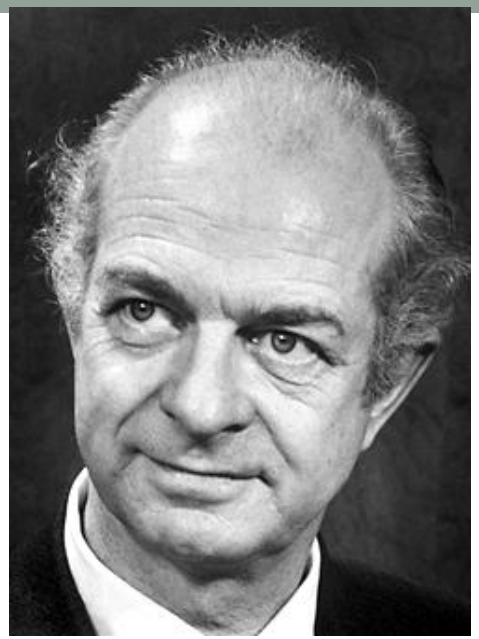
- Overview
- Molecular Orbital Theory (1927)
- Types of bonding
  - σ, π bonds
  - Bonding, antibonding, nonbonding
  - Bond order
- Diatomic Molecules
- Valence Bond Theory (1927)
  - Hybridization (1930)

Why Hybridize? (1) CH_4



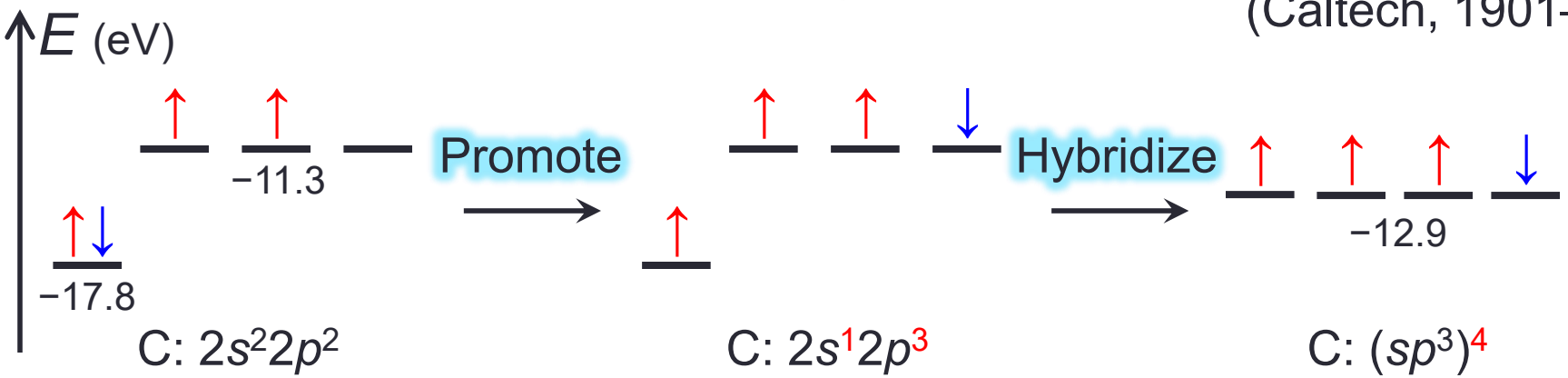
Pauling's Answer





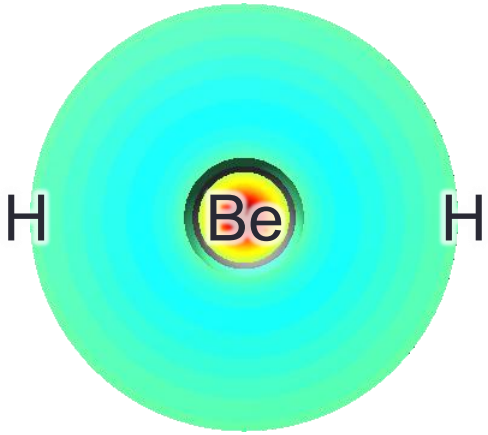
Linus Pauling
(Caltech, 1901–1994)

Or better,

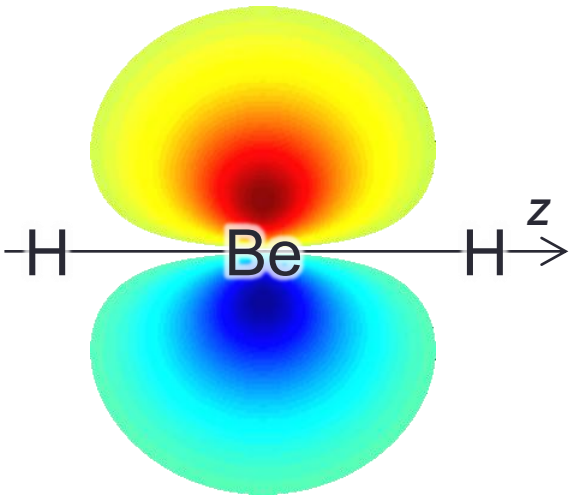


+ 4× H $1s^1$ ⟶ CH_4

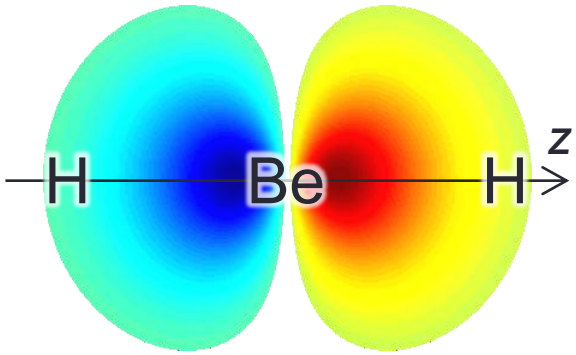
Why Hybridize? (2) BeH_2



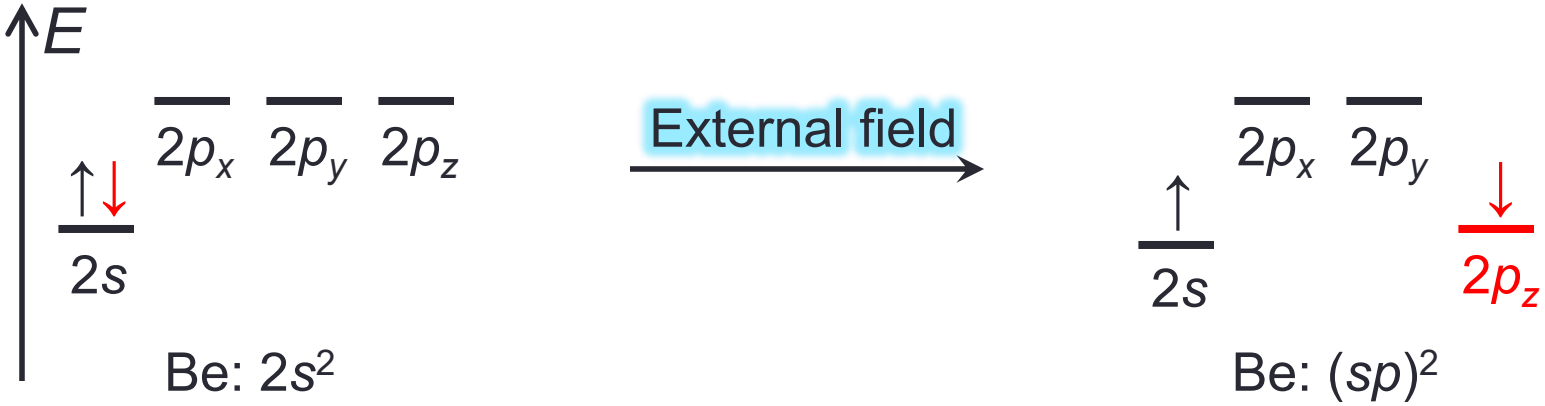
$2s$



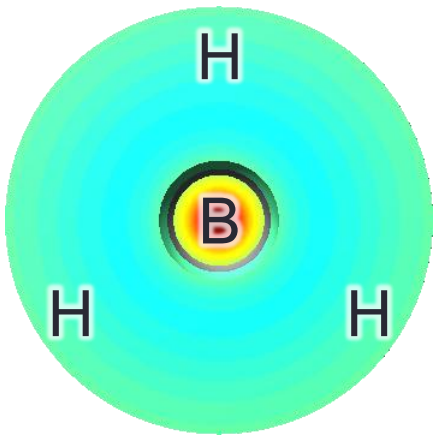
$2p_x$, $2p_y$



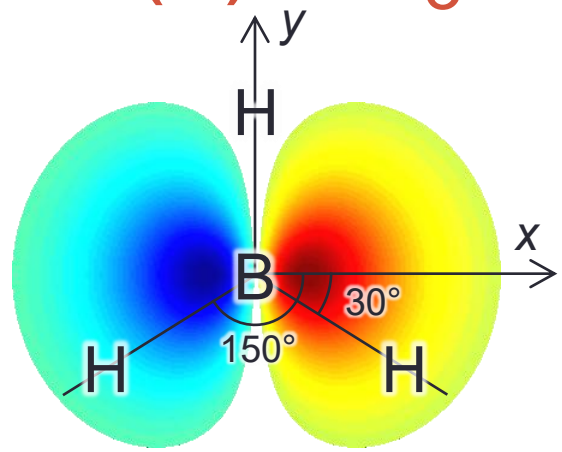
$2p_z$



Why Hybridize? (2) BH_3

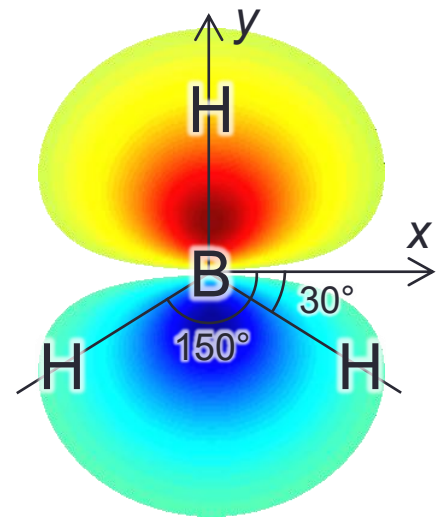


$2s$



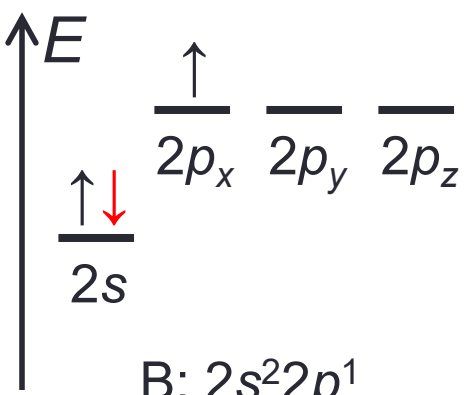
$2p_x$

$\cos^2(30°) + \cos^2(150°) = 3/2$



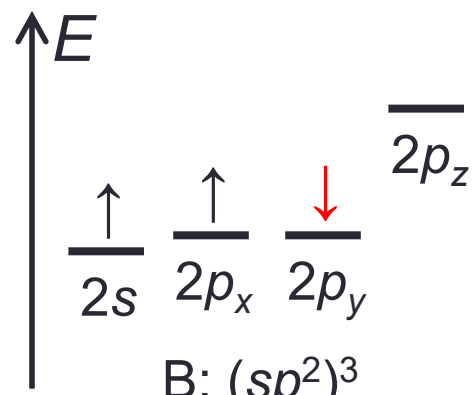
$2p_y$

$\sin^2(30°) + \sin^2(150°) + \sin^2(-90°) = 3/2$



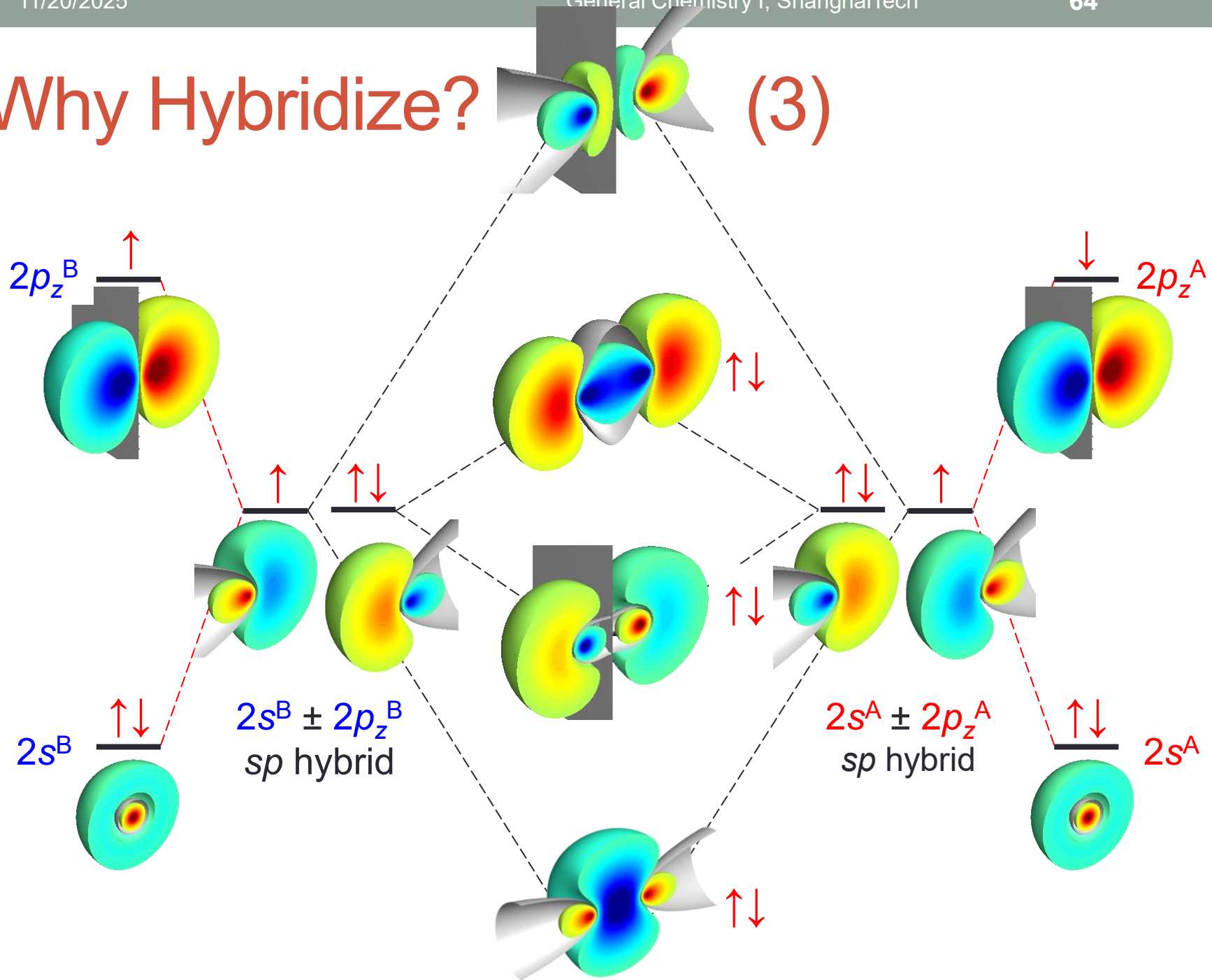
B: $2s^2 2p^1$

External field →

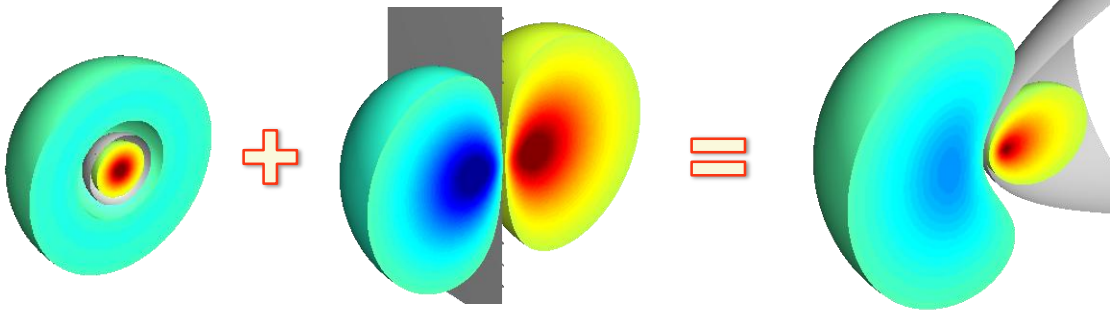


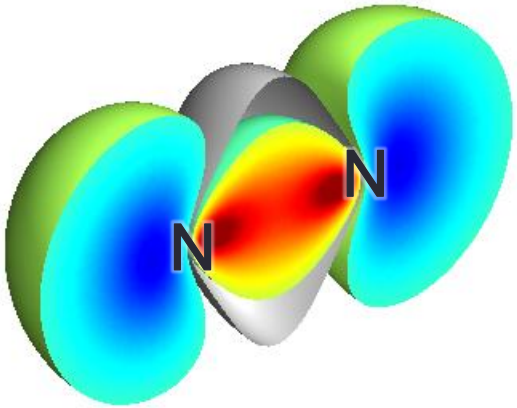
B: $(sp^2)^3$

Why Hybridize? (3)

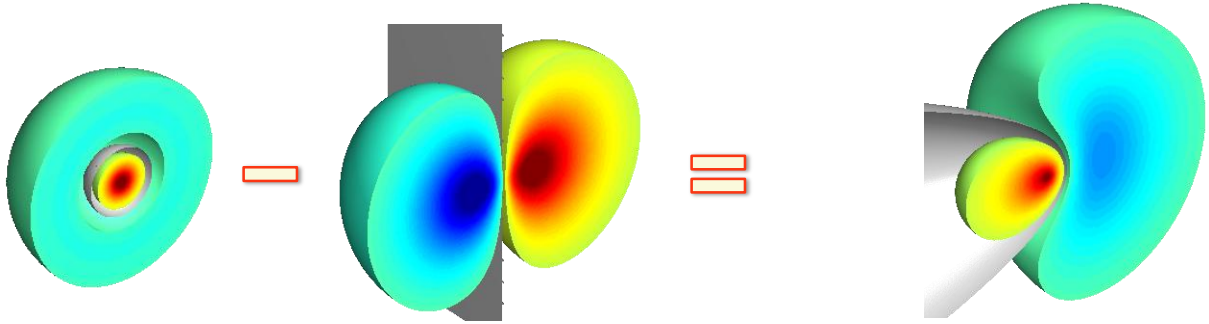


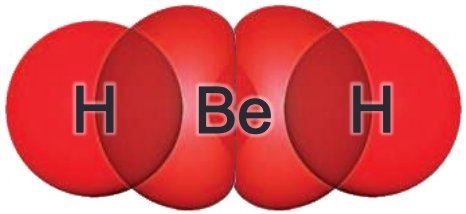
sp Hybridization



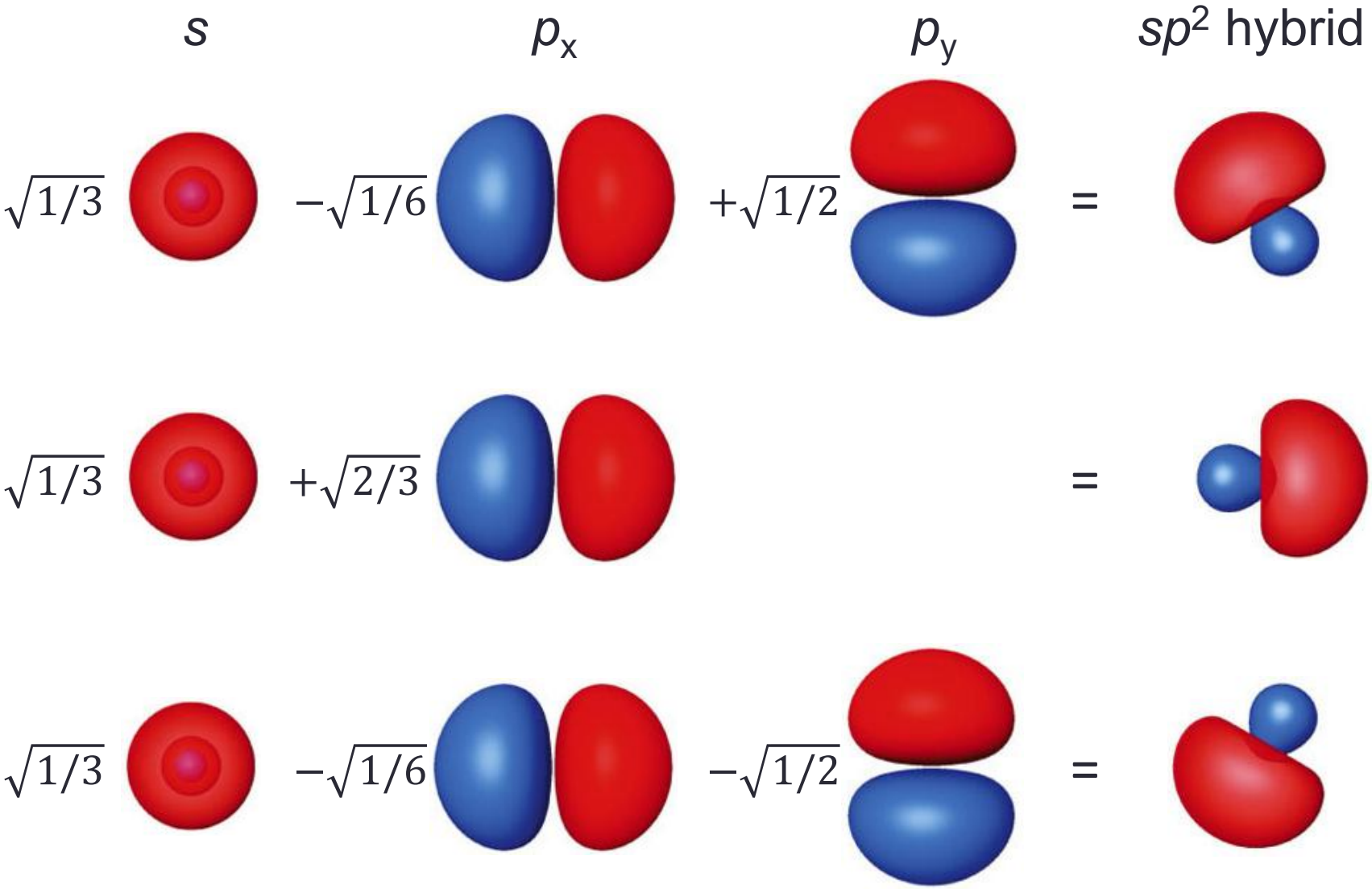


$\frac{1}{\sqrt{2}}s$ $\frac{1}{\sqrt{2}}p_z$ *sp* hybrid

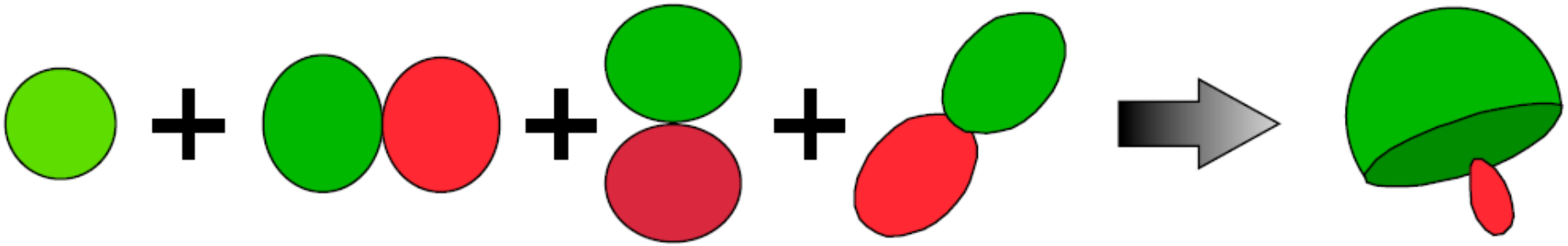




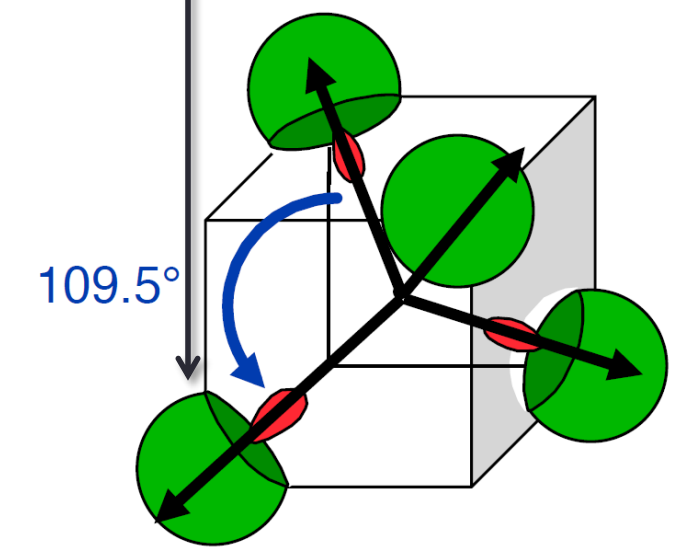
sp^2 Hybridization



sp^3 Hybridization

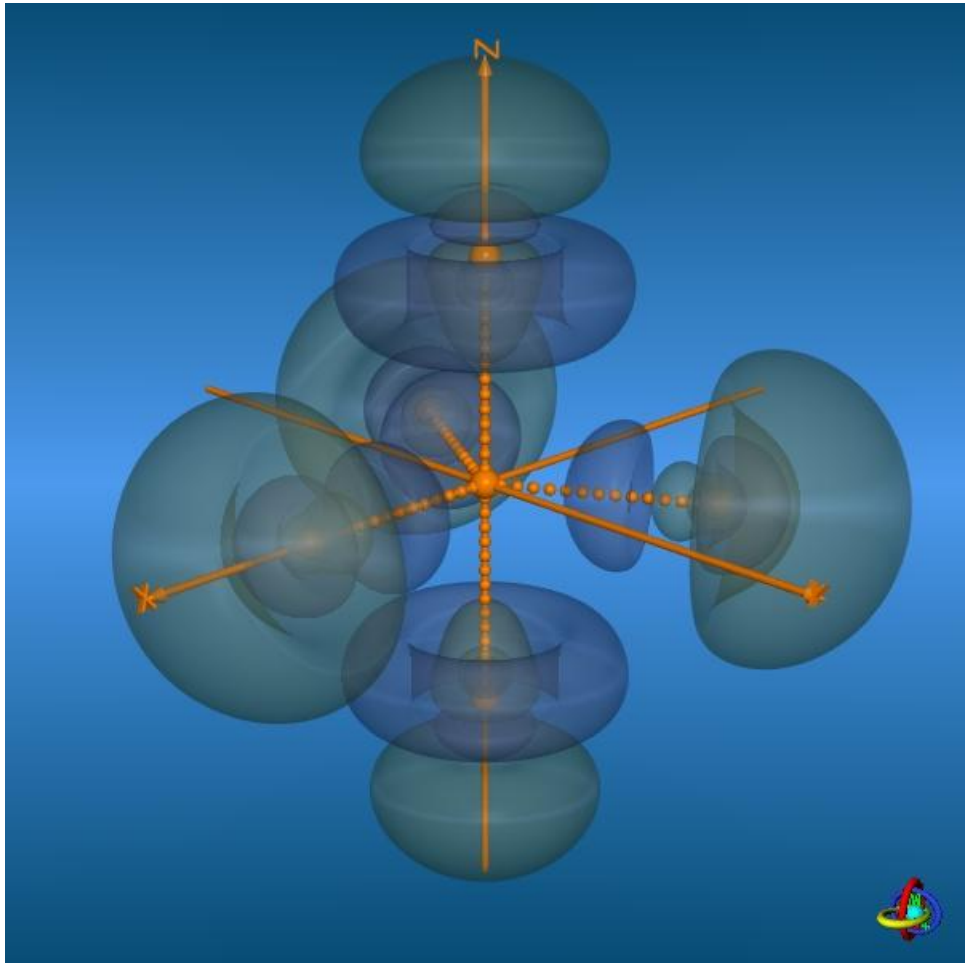


$\sqrt{1/4}$	$\sqrt{1/4}$	$\sqrt{1/4}$	$\sqrt{1/4}$
$\sqrt{1/4}$	$\sqrt{1/4}$	$-\sqrt{1/4}$	$-\sqrt{1/4}$
$\sqrt{1/4}$	$-\sqrt{1/4}$	$\sqrt{1/4}$	$-\sqrt{1/4}$
$\sqrt{1/4}$	$-\sqrt{1/4}$	$-\sqrt{1/4}$	$\sqrt{1/4}$

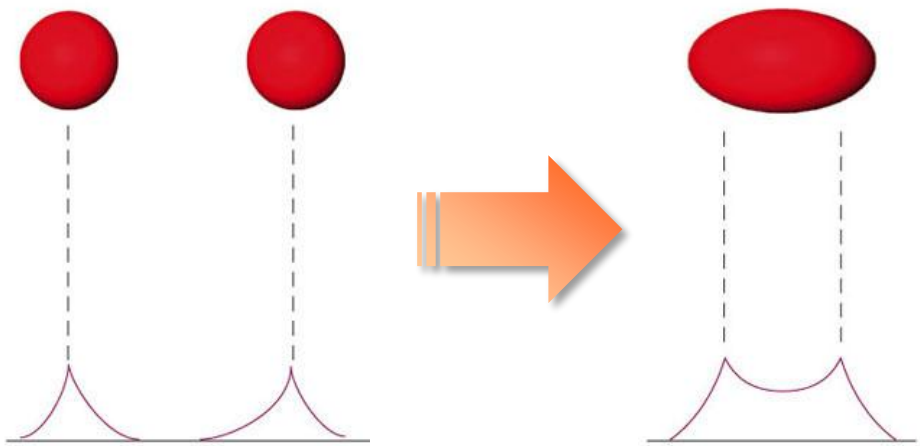


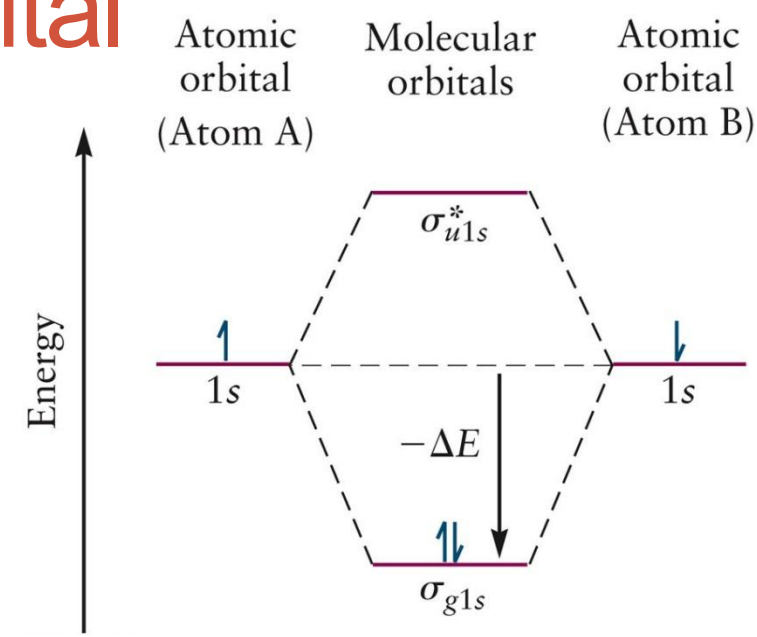
sp^3d Hybridization

s, p_x, p_y, p_z, d_{z2}



Summary: Bonding Orbital





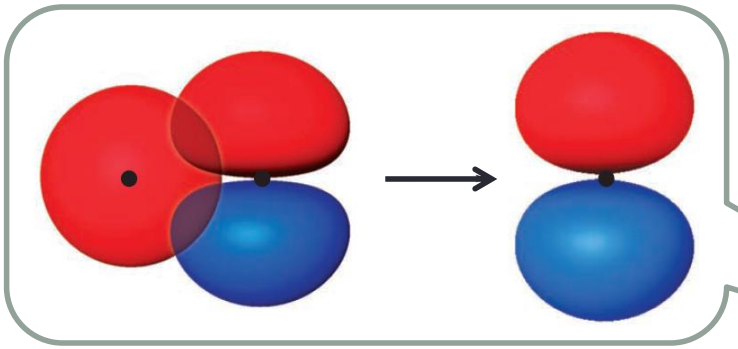
VB: Orbital overlap

MO: Bonding orbital

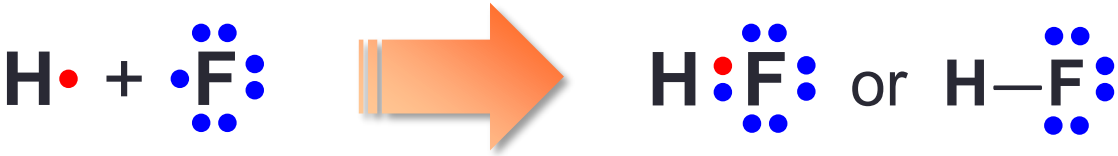
H• + •H → H:H or H×H or H—H

Lewis: Shared electron pair

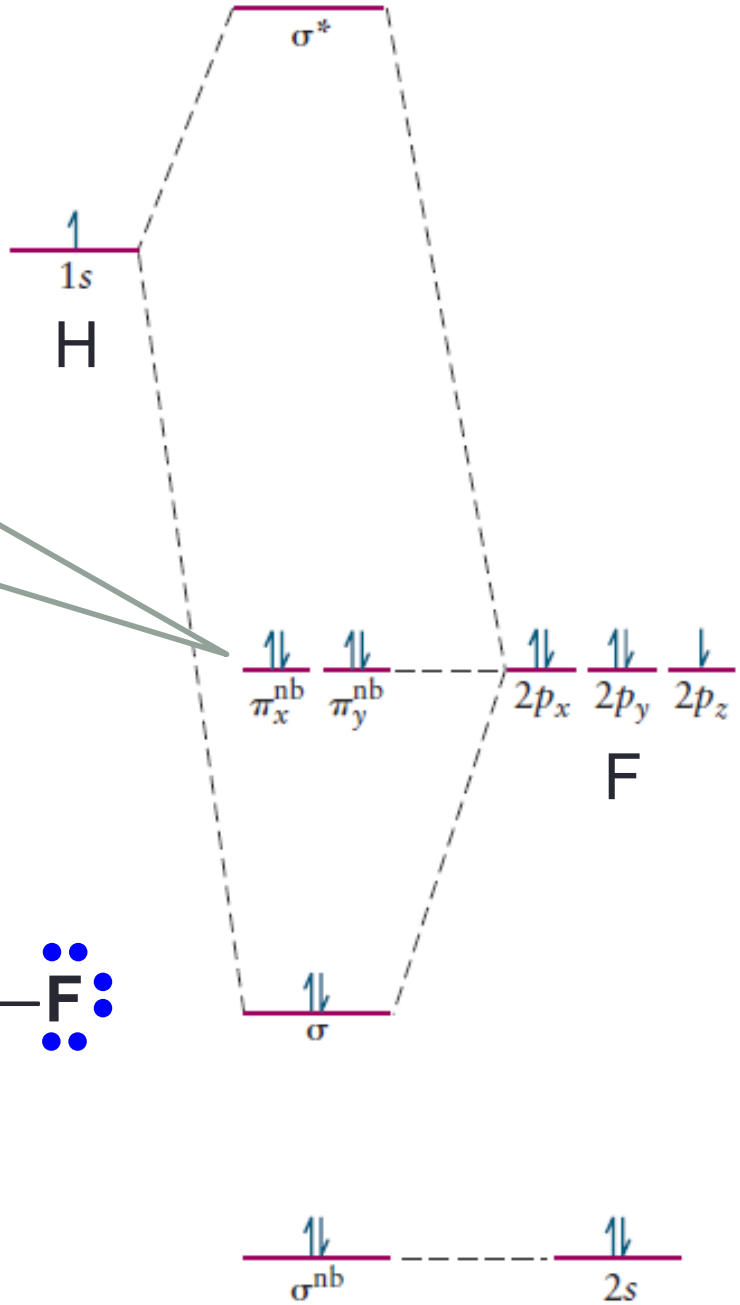
Summary: Lone Pair (1)



MO: Nonbonding orbital

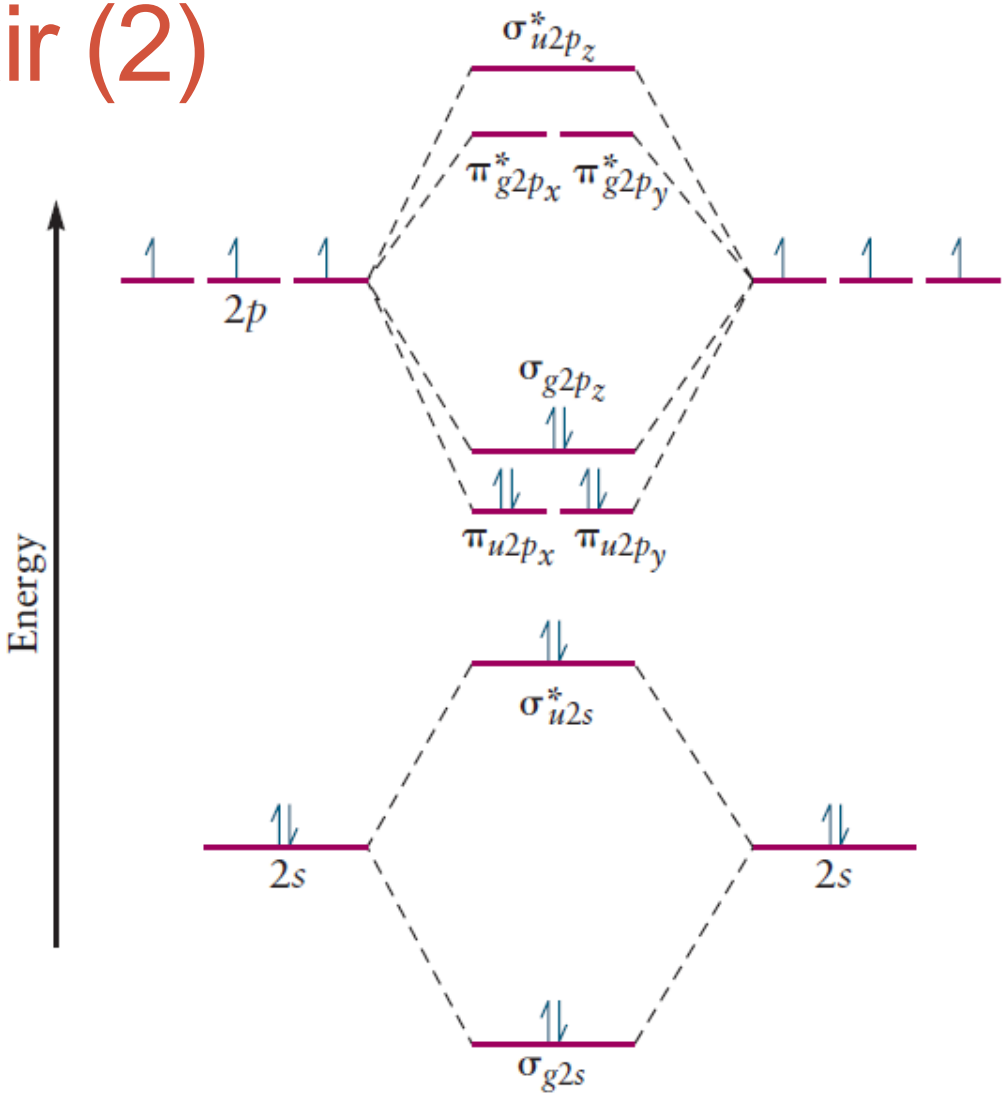


Lewis: Lone electron pair



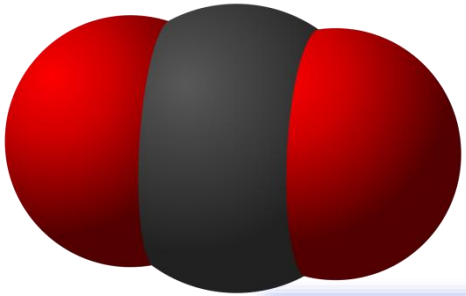
Summary: Lone Pair (2)

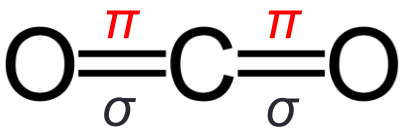
Lewis:
2 lone electron pairs

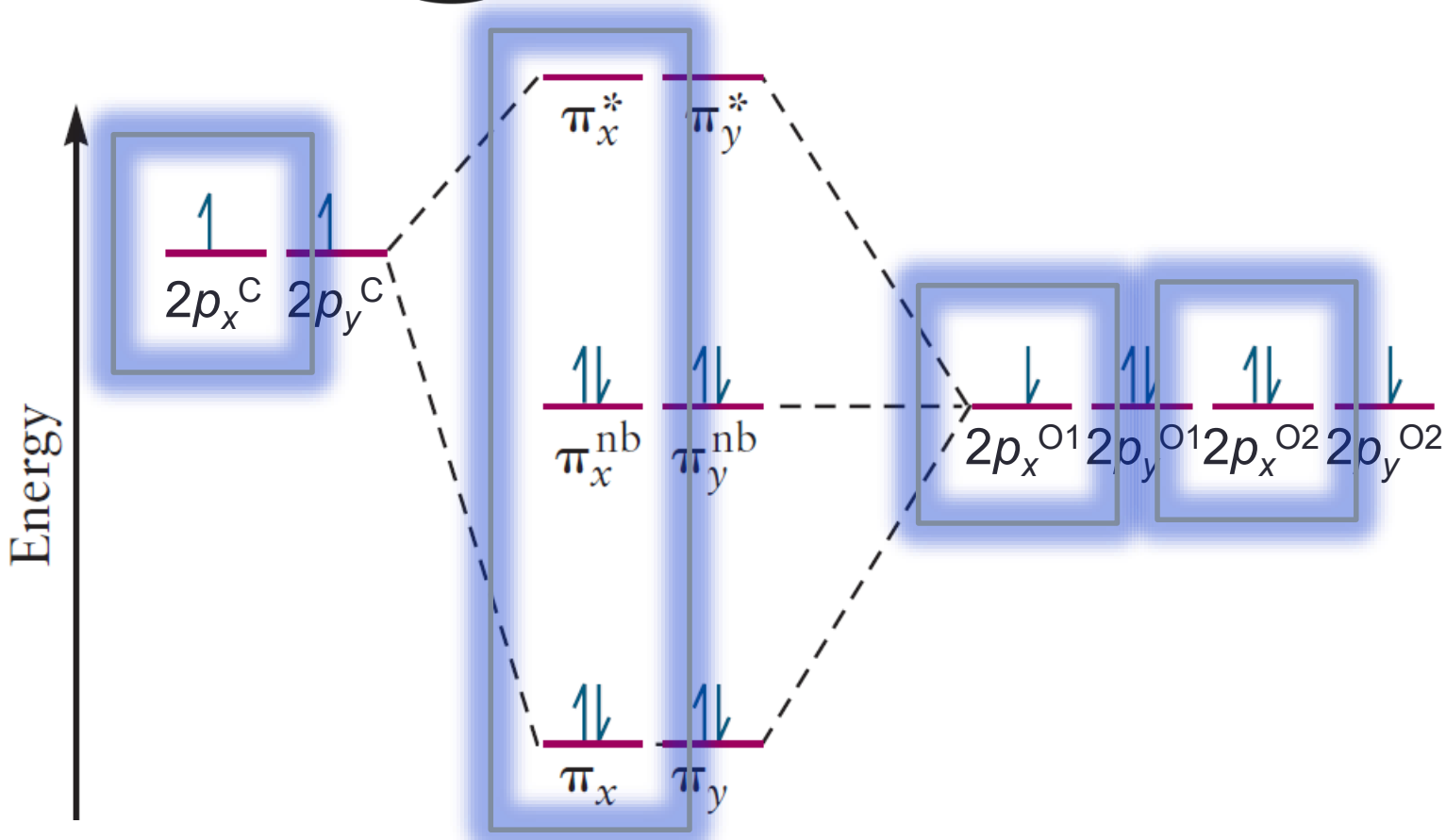


MO: Bonding + antibonding orbitals

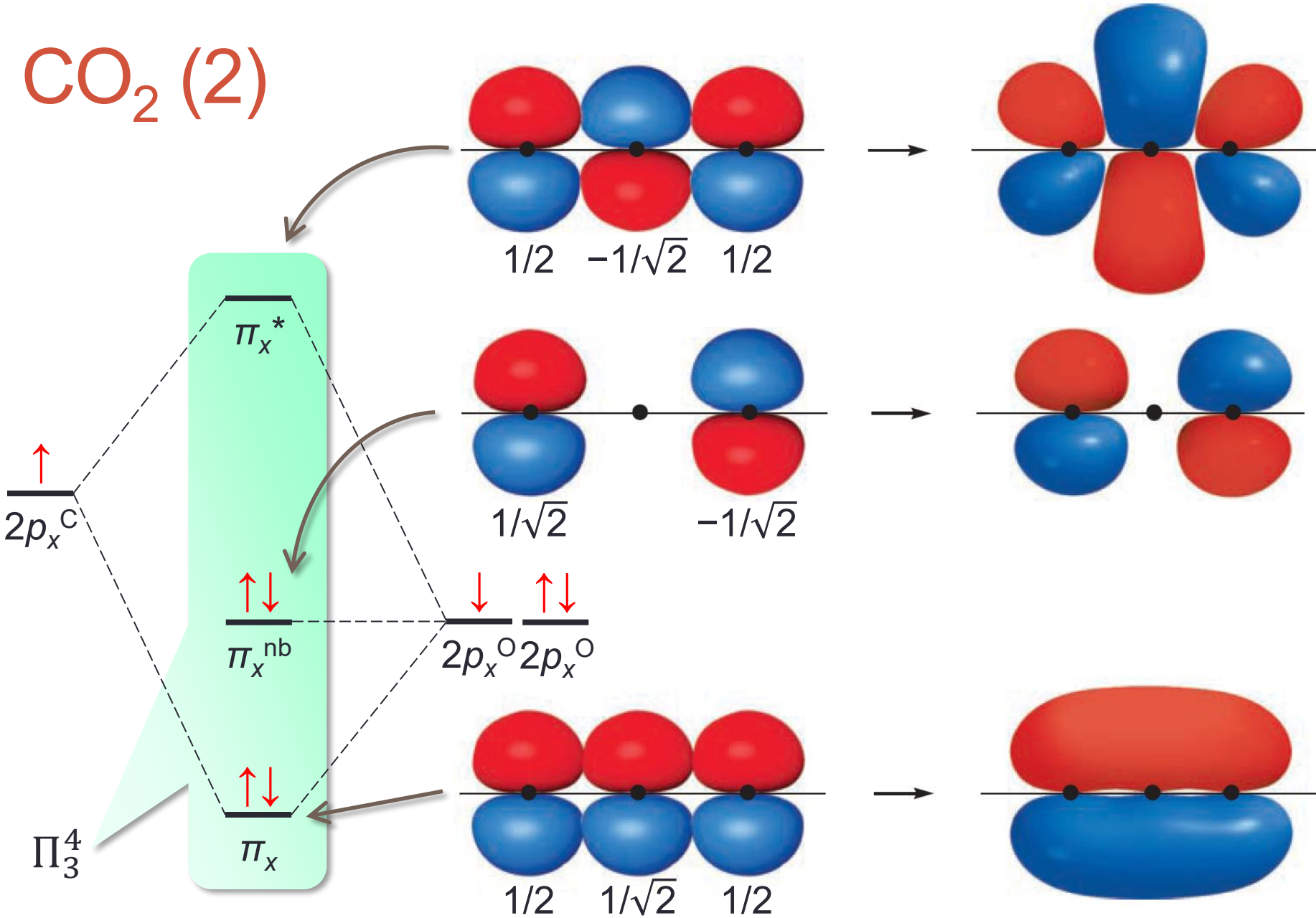
Triatomic Molecules: CO_2 (1)

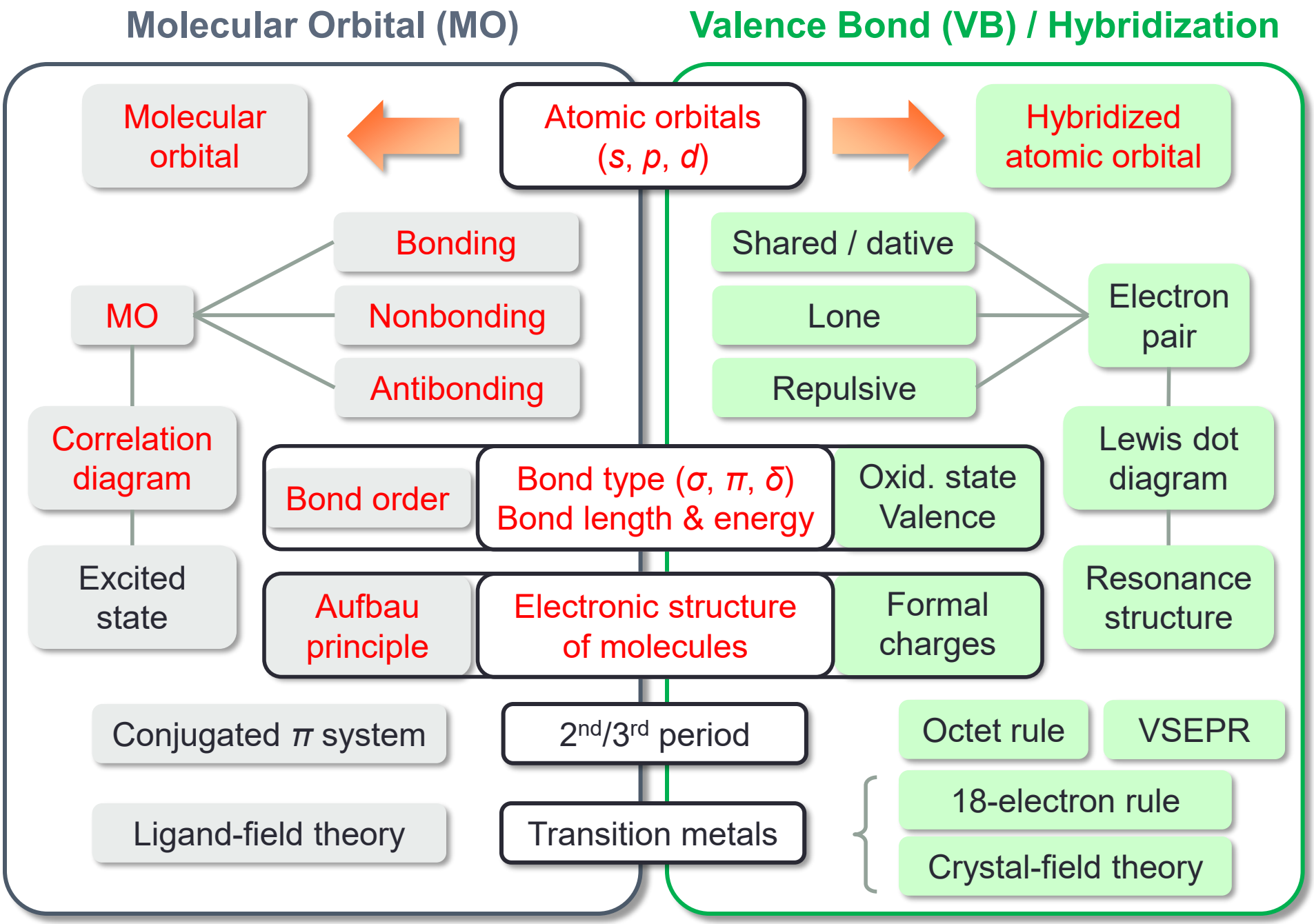






CO_2 (2)


π_x^*
$2p_x^C$
π_x^{nb}
$2p_x^O$ $2p_x^O$
π_x
Π_3^4
1/2 $-1/\sqrt{2}$ 1/2
$1/\sqrt{2}$ $-1/\sqrt{2}$
1/2 $1/\sqrt{2}$ 1/2


Molecular Orbital (MO)
Valence Bond (VB) / Hybridization
Molecular orbital
Atomic orbitals (s, p, d)
Hybridized atomic orbital
Bonding
MO
Nonbonding
Antibonding
Shared / dative
Lone
Repulsive
Electron pair
Correlation diagram
Excited state
Bond order
Bond type (σ, π, δ)
Bond length & energy
Oxid. state
Valence
Lewis dot diagram
Resonance structure
Aufbau principle
Electronic structure of molecules
Formal charges
Conjugated π system
2nd/3rd period
Octet rule
VSEPR
Ligand-field theory
Transition metals
18-electron rule
Crystal-field theory

Next Lecture Series: Rules of Bonding

Reading: OGB8 §3.10, §3.11